INCH - POUND

MIL-PRF-32335
14 July 2010

MILITARY PERFORMANCE SPECIFICATION
TRICON REFRIGERATED CONTAINER SYSTEM (TRCS)

This Performance Specification was prepared by US Army Natick Soldier Research Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers the Governments TriCon Refrigerated Container System (TRCS). The TRCS is made up of two principle components:

A Triple Container (TriCon) sized (8-foot wide by 6-foot 5 and half inches long by 8 foot high), International Organization of Standardization (ISO) compliant, insulated container (referred to as "container") and an integrated refrigeration/freezing/heating unit (referred to as "RU") per:

- Type I - electrically powered RU with full width double door container
- Type II - diesel/electric powered RU with full width double door container
- Type III - electrically powered RU with a single man door, recessed operating handle container

For Type I and II, the TRCS doors will be located on the side of the container (6.5 foot dimension) to allow full access inside when the container is coupled. The RU will be located on the side opposite to the door. For type III, a single man door with self closing pull style door handle is required.

When the container and the RU are combined, this will be referred to as the "system". The container will include certain added features called out in this performance specification. (i.e. interior tie downs, horizontal A or E track, air strip curtains, etc…).

1.2 Description. The TRCS described herein will be designed for the carriage of deep frozen, frozen, chilled and general cargo by road, rail, air, and sea (above or below deck), and will be suitable for the environmental conditions imposed by these modes of transport and suitable to be subjected to severe thermal shock. The TRCS will be designed to be coupled to other TRCS and/or dry freight TriCons using eight approved interconnectors (NSN 3040-01-387-4048 or equivalent). When three TRCS and/or dry freight TriCons are coupled the dimensions, tolerances, and diagonal differences of the three containers will meet the requirements for a 20-foot (1C) ISO container as specified in ISO 668.

Comments, suggestions or questions on this document should be addressed to US Army Research, Development and Engineering Command, Natick Soldier Center, AMSRD-NSC-CF-S, 15 Kansas St., Natick, MA 01760-5018, or emailed to Robert.Bernazzani@natick.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 4110

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The TRCS will comply with all applicable requirements of ISO 1496/2, 668, 1161, 6346, 49 CFR Chapter 4 parts 450-453, and Customs Convention. Unless otherwise specified herein, the TRCS container will comply with the requirements of ISO 1496-1 as applicable. The TRCS will be new and unused, waterproof, non-collapsible, of a permanent character and suitable for repeated use and repairable.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or those recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents, verifications and certifications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government Documents.

2.2.1 Standard. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (ASSIST) and supplement thereto, cited in the solicitation.

STANDARD

DEPARTMENT OF DEFENSE

MIL-STD-810	Environmental Engineering Considerations and Laboratory Tests
MIL-HDBK-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-DTL-22992	Connectors, Plugs and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type
MIL-STD-209	Interface Standard for Lifting and Tie-down Provisions
MIL-STD-889	Dissimilar Metals
MIL-STD-913	Requirements for the Certification of Sling Loaded Military Equipment for External Transportation by Department of Defense Helicopters.
MIL-STD-1366	Interface Standard for Transportability Criteria
MIL-STD-1474	Noise Limits
MIL-STD-38784	Standard Practice for Manual, Technical

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MIL-STD-40051 Preparation of Digital Technical Information

MIL-STD-461 Measurement of Electromagnetic Interference Characteristics

MIL-STD-882 Standard Practice for System Safety

FEDERAL STANDARDS

FED-STD-595/Color Chips 33446 and 34094 Colors Used in Government Procurement

(Unless otherwise indicated, copies of the above specifications and standards are available from Defense Automated Printing Service, Bldg. 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094, or https://assist.daps.dla.mil).

DEFENSE REGULATION

AR 70-38 Army Regulation: Research, Development, Test and Evaluation of Materiel for Extreme Climatic Conditions

(Unless otherwise indicated, copies of the above Army Regulation is available from http://www.army.mil/usapa/epubs/70_Series_Collection_1.html)

2.2.2 Other Government Documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DEPARTMENT OF TRANSPORTATION (DOT)

Code of Federal Regulations (CFR)

Title 49 CFR – Transportation, Parts 450-453

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 20402 or www.dot.gov).

TECHNICAL BULLETINS

TB-43-0214 Standards for Marking DoD Containers

TB-MED-530 Food Service Sanitation

(Application for electronic technical bulletins can be obtained at the following web site address www.army.mil/usapa/med/index.html).

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PRODUCT MANAGER – MOBILE ELECTRIC POWER TECHNICAL DATA

Medium Set	Technical Data
Large Set	Technical Data

(Application for electronic technical data can be obtained at the following web site address httpss://www.pm-mep.army.mil/).

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents, which are DoD adopted, are those listed in the issue of the ASSIST cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the ASSIST are the issues of the documents cited in the solicitation (see 6.2).

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 668	Series 1 Freight Containers - Classification, Dimensions and Ratings
ISO 830	Freight Containers - Terminology
ISO 1161	Series 1 Freight Containers - Corner Fittings – Specifications
ISO 1496-1	Series 1 Freight Containers - Specification and Testing - Part 1: General Cargo Containers for General Purpose
ISO 1496-2	Series 1 Freight Containers - Specification and Testing - Part 2: Thermal Containers
ISO 6346	Freight Containers - Coding, Identification and Marking

CSC, International Convention for Safe Containers

(Application for copies should be addressed to the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036, or www.iso.org).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F883	Standard Performance Specification for Padlocks
ASTM B117	Operating Salt Spray (Fog) Apparatus
ASTM-E-1976	Standard Specification for Shelters, Tactical Non-expandable
ASTM D 3359	Standard Test Method for Measuring Adhesion by Tape Test.
ASTM D 4138	Standard Test Method for Measurement of Dry Film Thickness of Protective Coating Systems by Destructive Means.

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(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-3959, or www.astm.org).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C101.1 Leakage Current for Appliances

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018, or www.ansi.org).

AIR CONDITIONING, HEATING, AND REFRIGERATION INSTITUTE

AHRI 1110 Performance Rating of Mechanical Transport Refrigeration Units

(Application for copies should be addressed to Air-Conditioning, Heating, and Refrigeration Institute (AHRI) 2111Wilson Blvd, Suite 500 Arlington, VA 22201 or http://www.ahrinet.org)

NATIONAL SANITATION FOUNDATION (NSF) INTERNATIONAL

Standard/ANSI No. 2 - Food Service Equipment

Standard No. 2 C-3 - Food Service, Special Equipment and/or Devices

Standard/ANSI No. 7 - Food Service Refrigerators and Storage Freezers

(Application for copies should be addressed to NSF International, P. O. Box 130141, Ann Arbor, MI 483113-0140, or www.nsf.gov).

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

Standard No. 70 - National Electrical Code

NFPA 77 – Recommended Practices on Static Electricity

(Application for copies should be addressed to the National Fire Protection Association, One Battery March Park, Quincy, MA 02269-9101, or www.nfpa.org).

OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION STANDARDS (OSHA)

OSHA 29 CFR 1910 - Occupational Safety and Health Administration Standards (Application for copies should be addressed to the Department of Transportation, Superintendent of Documents, Government Printing Office, Washington, DC 20402, or www.osha.gov)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Qualification Standard for Welding and Brazing Procedures, Welders, Brazers and Welding and Brazing Operators.

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(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th St., New York, NY 10017, or www.asme.org).

AMERICAN WELDING SOCIETY (AWS)

B2.2	-	Standard for Brazing Procedure and Performance Qualification.
D1.1	-	Structural Welding Code - Steel.
D1.2	-	Structural Welding Code - Aluminum.

(Application for copies should be addressed to the American Welding Society, 550 NW LeJeune Road, PO Box 351040, Miami, FL 33135, or www.aws.org).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Standards Publication No. 250 - Enclosures for Equipment, Electrical (1000V Maximum)

(Application for copies should be addressed to the National Electrical Manufacturers Association, 1300 No. 17th Street, Suite 1847, Rosalyn, VA 22209, or www.nema.org).

UNDERWRITERS LABORATORIES INC. (UL)

Standard 207 – Refrigerant Containing Components and Accessories, Non-electrical

Standard 353 - Limit Controls

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062-3096, or www.ul.com).

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS8090 Mobility, Towed Aerospace Ground Equipment, General Requirements for

(Application for copies should be addressed to the SAE, The Engineering Society for the Advancing Mobility Land Sea Air and Space International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, or www.sae.org).

2.4 Order of Precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Operating Requirements. The TRCS shall satisfy the following minimal functional, operational, and performance requirements.

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3.2.1 Air Leakage (ISO 1496-2). The overall air leakage of the TRCS shall not exceed the following:

Type I, II and III	7 m^3/hr at 25.4 ± 1mm (1 inch) of water pressure gauge inside (with the refrigeration unit in place)

3.2.2 Heat Leakage. The overall heat transfer rate of the TRCS shall be minimized and shall not exceed the following: This includes the heat transfer rate through the refrigeration unit.

Type I, Type II, Type III	11.0 Kcal/°C/hr with a mean wall temperature of 293K (20°C) (with the refrigeration unit in place)

A high quality insulating material shall be utilized and conductive heat paths eliminated or reduced to achieve a low overall heat leakage coefficient.

3.2.3 Airflow and Air Circulation. The interior of the TRCS shall be designed to ensure airflow in and around the cargo to insure uniform distribution of conditioned air. Enough space shall be provided for proper circulation of the air without any sort of blockage by contents and other products stored in the TRCS. The TRCS shall have permanent markings located inside the container to establish where cargo cannot be stored, so as to facilitate the proper airflow. A protective cage shall be provided over the evaporator and condenser coils to prevent hand/finger injuries, fin damage and prevent the users from storing items at those locations.

3.2.4 Refrigeration Capacity. The TRCS shall provide sufficient refrigeration capacity to maintain internal set point temperatures and cargo temperatures in all ambient conditions listed herein. The refrigeration units shall perform capacity tests at 0°F and 35°F (set point temperatures) inside the TRCS with ambient temperatures of 100°F at 69% Relative Humidity (RH) and 125°F (+0/-3°F) at 3% RH. The capacity tests shall use the methods described in ARI-1110, (Air Conditioning & Refrigeration Institute) “Standard for Mechanical Transport Refrigeration Units”. The unit shall be sized properly for above conditions and shall not be too large as to cause short cycling of the refrigeration unit. The refrigeration unit shall have a minimum of 50% extra net cooling capacity above the overall heat leakage coefficient of the TRCS insulated container at 100°F and maintain set point temperatures with a (+/-3°F) tolerance in all cooling and heating modes. The refrigeration unit shall pull down the interior temperature (both wall and air temperature) of the container from 125°F to 0°F in less than 9 hours. The refrigeration unit shall also be energy efficient and operate properly in all military transportation and storage operational scenarios.

* The 0°F and 35°F set point thermostat setting temperatures are identified as the return air temperature entering the evaporator and the return ambient air temperatures of 100°F at 69% RH and 125° (+0/-3°F) at 3% RH entering the condenser.

3.2.5 Heating Capacity. The TRCS shall be capable of maintaining temperature set points (of both temperature sensitive cargo and internal air temperatures) through the addition of heat when ambient conditions cause the interior temperature to drop below the set point. The TRCS shall be designed and capable of providing sufficient heating capacity, to maintain 60°F, at ambient temperatures as low as -35°F.

3.2.6 Moisture Content and Refrigerant Leakage. The refrigeration system shall be evacuated and charged to the RU manufacturer’s specifications. RU and refrigerant-containing components shall be

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gastight. The Refrigeration unit system shall have a refrigerant moisture indicator that is visible during operation, maintenance and repair operations and a refrigerant filter dryer accessible to the maintainer.

3.2.7 Defrost. The RU shall have a defrost cycle terminated by temperature and automatically initiated on demand (when ice builds up on the evaporator coil) or shall automatically defrost the evaporator coil by measuring the difference between the return air temperature and the saturated inlet air temperature. Alternatively, the defrost can be initiated by a time control adjustable from 2 to 12 hour intervals and a user selected timed defrost termination. Hot Gas Bypass or an all electric defrost is acceptable. The defrost system shall have provisions for manual defrost (override) initiation and controls easily accessible by the operator along with the other common controls. The defrost system shall have the capacity to remove condensate build up on the evaporator coil and return to normal cooling modes. Refrigeration units shall have a drainage system for the collection and discharge of condensate to the outside of the insulated container without pooling of the fluids in the lines. Condensate drain lines shall prevent the infiltration of insects into the lines. The drain line assembly shall utilize commercially available components that are easily disconnected and reinstalled during maintenance and removal of the RU from the IC. The condensate drain assembly shall not damage any rations, equipment, or stored items located in the container. The condensate drain hoses and associated fittings, whether internally or externally mounted, shall be physically protected from damage and from freezing temperatures. The RU shall prevent freezing of condensate in the drain pan and condensate drain hose assemblies.

3.2.8 Safety. The TRCS shall be safe for the intended purpose and shall minimize flammability, operational, electrical, and toxic personnel hazards. Hazards shall be considered up front in the design phase and eliminated or minimized to the maximum extent possible. Potential, operational and personnel hazards shall be identified to the end user with permanent stenciling, non-reflective placards or labels, and in the Technical Manual. Yellow and black “Caution Tape” shall be permanently affixed to the exposed surfaces of the evaporator to prevent possible head injuries. All controls shall be permanently labeled and traceable to the Technical Manual. All Electrical and Electronics used in the TRCS shall be permanently marked for identification for proper and safe field usage. The TRCS shall be free from sharp edges, pinching hazards, etc.

Note: the following safety requirements shall also be met:

General. The RU shall be designed so that under all conditions of normal use (installation, operation, and maintenance) and under a likely fault condition (including human error), it protects against the risk of electric shock and other hazards. All hazards shall be eliminated or reduced to the lowest risk level practicable using methods in the following order of precedence: design; incorporation of safety devices; incorporation of warning devices; and procedures/training. Catastrophic or critical hazards shall not rely solely on warnings, cautions, or procedures/training for control of risk. All safety hazards not eliminated through design shall be addressed in the appropriate technical manuals. Information regarding hazard-avoiding procedures and safety warning labels on equipment shall be included in all manuals.
Electrical safety. The following shall be incorporated into the RU design:
Dangerous voltages and currents. Operators shall not be exposed to components with voltages exceeding 30V. The operator shall not be exposed to any stored energy shock at the disconnecting

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means IAW UL 1950, paragraph 2.1.10. Protection shall be provided to personnel during maintenance and repair to prevent unintentional contact with voltages exceeding 70V. Capacitors shall be provided with an automatic discharge paths when power is removed.
Grounding. Hinges and slides shall not be relied upon as the sole means of grounding. Bleed resistors shall be used where resistor networks discharge capacitors, or act as voltage dividers (e.g., between test point and ground). Equipment leakage current to ground shall not exceed 3.5 mA, when tested IAW ANSI C101.1. Where leakage current exceeds this value, redundant equipment grounding paths shall be provided in the design.
Circuit protection. A resettable circuit breaker shall be installed in/on the RU which, when activated, will cut off all power into the RU. Fuses are not acceptable on the main incoming power. Resettable circuit breakers are preferred on internal electrical components. Replaceable fuses are acceptable on internal electrical components only.
Connectors. All electrically hot contacts shall be socket contacts or ring terminals. All electrical connectors that must be mated in the field (except for the main power connector) shall be quick disconnecting types. All multi-pin connectors shall be keyed to prevent incorrect mating or insertion. All protective covers shall be captive. All fuses, relays, and circuit breakers shall be labeled to identify which component they control. The TRCS shall be fitted with a Grounding lug for 12AWG wire.
Mechanical safety. The following shall be incorporated into the RU design: a. The equipment shall provide maximum access and safety to personnel during installation, operation, and maintenance. b. The RU shall be designed so that it can be moved, handled, and lifted safely. c. Any pressure relief valve must vent in a safe location and manner. d. All rotating or reciprocating parts and parts subject to high operational temperatures that are of such nature or so located as to be a hazard to operating personnel shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.2.9 Uncontrolled Hazards. The TRCS shall present no uncontrolled hazards to operators or pose any potential damage to Government equipment. Fail-safe design shall be used for critical safety/health function components or subsystems. Safeguards shall be designed into the system, providing a clear indication of any unauthorized tampering that has taken place with the system.

3.2.10 Floor drainage. The TRCS shall be watertight and capable of being drained thoroughly through the floor drains without any blockage. The TRCS shall have automatic floor drains, which provide complete and total drainage of the container. The drains shall open automatically when subjected to pressures above normal internal pressure. The drain openings shall be recessed below the bottom surface of the floor. A minimum of four (4) floor drains shall be provided. The drains shall allow the user access to perform maintenance procedures. The drains shall facilitate quick draining (less than one minute) during cleaning operations. The operation of the drains shall not be negatively affected by sand that could be tracked or blown into the TRCS container in sandy/dusty environments.

3.2.11 Pressure Relief. The TRCS container shall be provided with a "automatic pressure relief mechanism" to equalize either positive or negative pressures inside the container to support daily ground and in-flight use and must meet the rapid decompression requirement of MIL-STD-810, Method

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500.0, procedure I. For type I and II only, an emergency/air pressure relief hatch in the right hand door of the insulated container is required to equalize either positive or negative pressures inside the container to support daily ground and in-flight use and must meet the rapid decompression requirement of MIL-STD-810, Method 500.0, procedure I. The hatch must be operable from the interior of the container and shall prevent air/water intrusion when in the closed position (See Section 3.2.18).

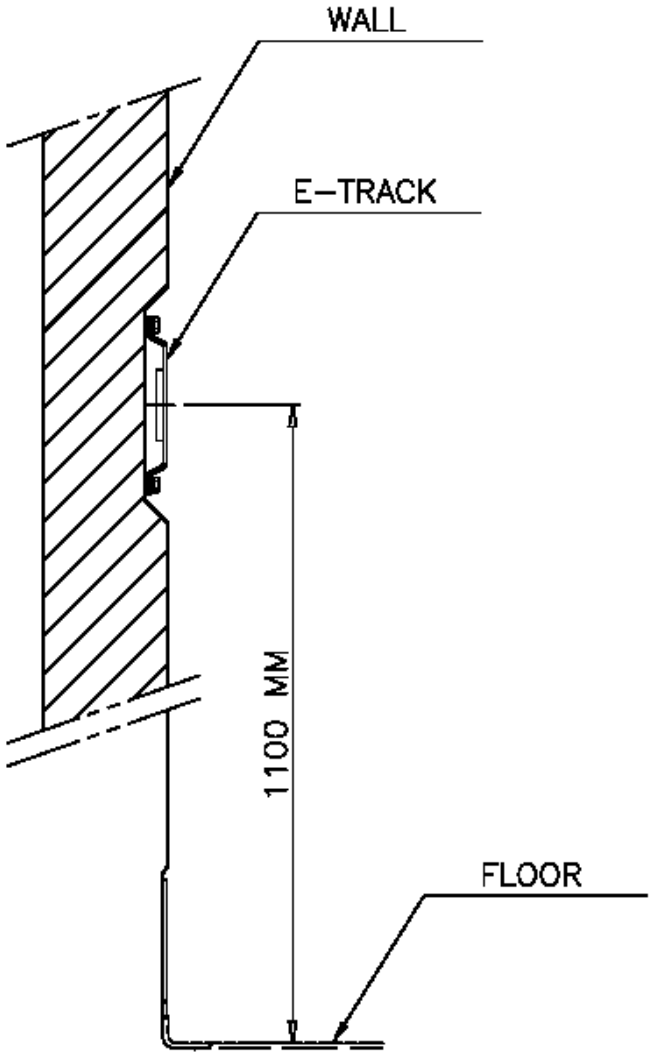
3.2.12 High Pressure Cutouts. The RU shall have an appropriate re-settable high pressure cutout switch to protect the RU from damage. The RU shall have a mechanism to relieve pressure in a safe location and manner. This mechanism shall activate if the system pressure reaches the design load limits and/or the limits set by the compressor manufacturer. This mechanism is intended to prevent excessive pressure inside the system from causing catastrophic mechanical failure.

3.2.13 Heater High Limit Cutout. The RU shall have a heater high limit cutout device to protect the unit from damage. The cutout for the heater high limit device shall be set in accordance with commercial practices. This switch shall be designed to reset automatically. All materials used in the compartment where the heating elements are located shall be able to withstand the exposure to high temperatures without charring, becoming brittle, or otherwise becoming damaged during the period of time typical of a loss of airflow/over-temperature shutdown scenario.

3.2.14 Seams. All exterior TRCS seams shall be permanently sealed and not allow the intrusion of water and air. Internal seams shall be sealed with food safe sealant, where required.

3.2.15 Cargo Restraint System. A minimum of 12 tie-down points shall be provided in each TRCS container. The tie down points shall be fully compatible with special purpose tie downs straps (NSN 3990-01-304-3009) and/or (NSN 1670-00-725-1437). For Type I, there shall be four tie downs on each of the lower portions of the front and rear (8') walls of the container and two tie downs on the RU and door side walls/floors. For Type III, the two tie downs located on the door side shall be mounted on the lower portion of the walls. Any tie downs located on the floor shall be fully recessed. All tie-downs shall be capable of accepting one-inch wide metal strapping without tearing the strapping. Adequate clearance shall be provided at the tie-down for installation of the metal hook of adjustable web strapping. All tie-downs shall not create a direct thermal short to the outside of the container and shall withstand a minimum 1,000 lbs load at a 45-degree pull angle without permanent deformation. The tie-down provisions shall restrain the load without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. The tie downs shall also be corrosion resistant. Each TRCS shall be supplied with a minimum of seven (7) "special purpose straps ((web, tie down, NSN: 3990-01-304-3009, PN: FDC5770-5 (Cage Code: 98313)) placed inside the container. The interior of the TRCS container shall also be compatible with the mounting or and the rated loads for the shoring beam (NSN: 9540-01-491-3804). The interior of the container shall also be supplied with a minimum of two (2) A or E-tracks per the following illustration to meet the critical Army equipment pack out requirements.

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3.2.16 Vinyl Air Strip Curtains. The TRCS door openings shall have removable commercial grade, full door width, slide mounted, heavy-duty, vinyl, air strip curtains fitted just inside the cargo doors to prevent the loss of cooled air when the doors are open. Strip curtains shall block ambient air from entering the IC when the doors are in the open position. The strip curtains shall be easily installed, removed and re-installed as necessary in the field without special tools. The strip curtains shall not affect the operation of the doors and ingress/egress of personnel.

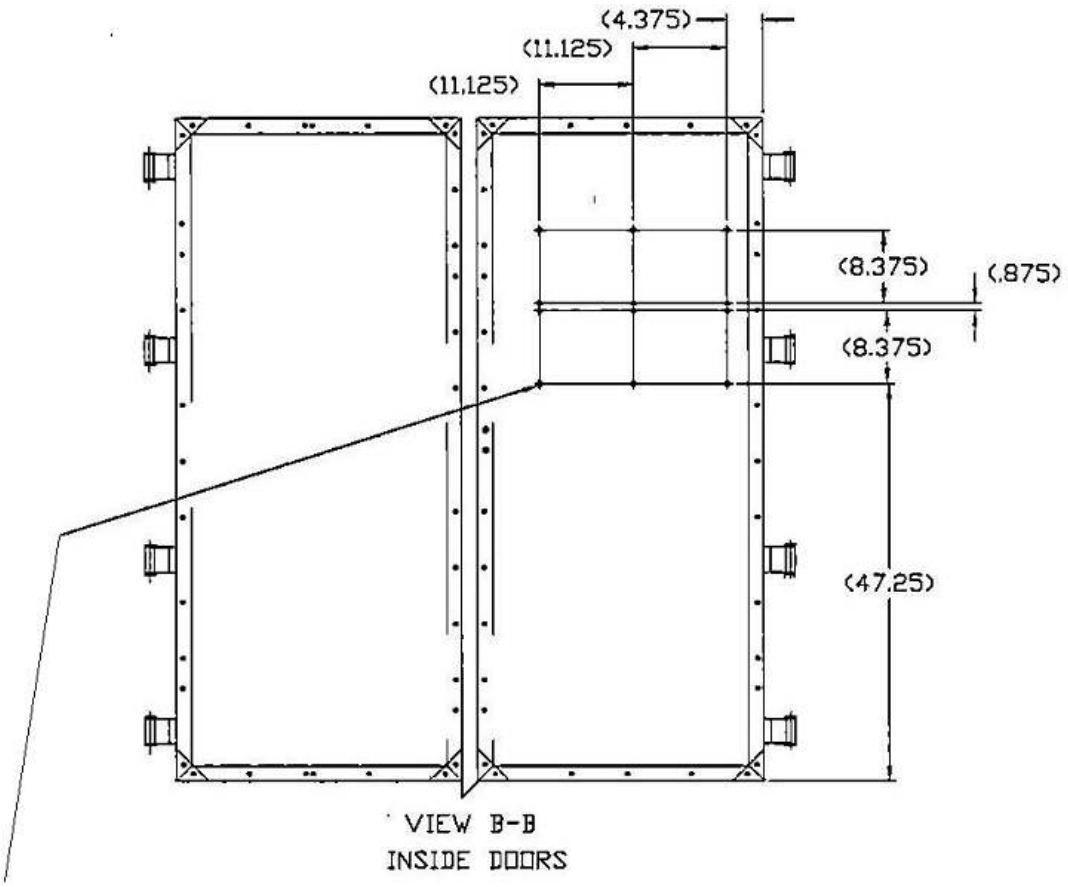
3.2.17 Technical Manual Holder. One or two waterproof and weatherproof, technical manual holders shall be securely fastened to the exterior within the ISO dimensional limits listed herein. The closing and latching devices of the holder shall be capable of repeated opening and closing, without tools and without compromising the sealing quality or causing any type of damage. The door of the manual holder shall be captive to the manual holder when opened. The interior dimensions of the holder shall be sized properly to contain the technical manual, with dimensions of not less than 12 inches long by 9 inches wide by 3 inches deep (or sufficient to house all required support documents). This includes, but may not be limited to, technical manuals, maintenance manuals, operator manuals, repair parts manuals

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and other required support documents. Permanent black lettering reading, “TECHNICAL MANUAL HOLDER”, shall be permanently stenciled on the exterior of the manual holder. Two manual holders located in the door section with a thickness not greater than 1.5 inches is acceptable in lieu of one manual holder.

3.2.18 Door Configurations. The exterior lip of the door gaskets shall provide leak free seal with the containers exterior surface when closed. An emergency release cargo door hardware shall be provided to permit a person to exit a locked TRCS from the inside. The doors shall be provided with a means to prevent door hardware from damaging the exterior container skin when opened. Type I and II doors shall be full opening and provide full access to the interior of the container. External hardware shall be protected from damage and shall not extend beyond the ISO clearance allowances. A means shall be provided to hold doors in full open position. On Type I and II doors, a small watertight and weatherproof hatch shall be provided on the right hand door that is locked and secured from the inside of the container to vent the container, when transported by air, in the case of rapid decompression, and to prevent a person locked inside the container from suffocation. The hatch location shall be identified to the user and visible from inside the container when the doors are closed from the exterior. Luminescent marking tape is allowable. A minimum of four (4) hinges, one (1) locking bar and one (1) grip handle shall be supplied per door for Type I, II or III containers. One interior door for Type I and II TRCS containers shall have blind rivnuts installed per the following illustrations (View B-B, Inside Doors) to mount information placards that display packing plan information for the container (dimensions in inches).

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INSTALL M5 BLIND RIVNUTS 12 X S.S.
REINFORCED PLATES AND SEALANT AS
REQUIRED, WITH MATING HARDWARE PROVIDED

3.2.19 Door Security. Each door shall be provided with minimum one heavy-duty, external handle operated cam-locking device with anti-rack provisions, which through lever type action aids in releasing the door seal from the doorframe. Each locking device handle must be capable of accepting a padlock or security seal. The handle and latch shall not extend beyond the ISO clearance allowances. Additional pull handles shall be provided to facilitate door opening. A means shall be provided to hold the doors in the full open position and shall be of a material, which will not scrape or chafe the container when the doors are closed. All moving parts of the door locking mechanism and door hinges shall be permanently lubricated or designed in such a way that lubrication may be performed during container maintenance checks. External hardware shall be protected from damage. The door and hinge assembly shall be capable of supporting a 200 lb. static load without degradation or damage to the door or door hardware and the door shall open, close and operate freely after such static load is applied. The door shall be easily opened by the 5% to 95% female/male warfighter. Each set of doors shall be equipped with a means to safely exit the container in the event that all of the doors are locked (to assume all possible locking configurations). This escape means shall have luminescent strips to direct an individual to the egress. All locking device handles shall be furnished with provisions for padlocking and customs sealing according to ASTM F883. No door hardware shall extend beyond the ISO clearance allowances or overall ISO dimensions.

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3.2.20 Anti-Pilferage Provisions. Hinge-pins and screws, bolts, and other fasteners used for securing the hinges and closing devices to the container, and for holding the essential parts of the sides, ends and roof, shall be welded or otherwise secured in such a manner as to prevent access to the interior of the container without leaving visible signs of tampering. Where such welding destroys the protective coating on the items being welded or on other container parts, the protective coating and the surrounding areas shall be restored to the original condition. All locking device handles shall be furnished with provisions for padlocking and customs sealing.

3.2.21 4-Hour Operation. The TRCS shall provide refrigeration, defrost, and heat as intended with suction and discharge pressures safely controlled and the proper and safe voltages and currents controlled for 4 hours.

3.2.22 Temperature Indicator. A primary temperature indicator shall be provided on the refrigeration unit controller. The temperature indicator shall be vapor tight and waterproof. The sensing element shall be located at the inlet side of the evaporator. The sensing tube shall be protected from damage during all modes of operational use and transportation. The temperature indicator shall be capable of providing readings for all temperature ranges encountered during operation of the TRCS unit as a field site refrigerator/freezer.

3.2.23 Temperature Control and Control Panel. The selected temperature setting of the RU shall be controlled automatically to provide selectable refrigerator, freezer or heating temperatures inside the insulated container. The temperature control shall allow incremental selection of internal normal refrigerator, freezer and heating temperatures from outside the container. The end user shall be able to set or adjust the set point temperature and the TRCS temperature control system shall be capable of controlling the temperatures in the interior of the container. Thermostats shall display, and be capable of being programmed, in both °C and °F. When set point temperatures below 20ºF are selected, the heating mode shall not operate. If the RU has a separate refrigeration and freezer modes, the heater shall not activate in freezer mode. The refrigeration unit shall be provided with an externally accessible watertight control panel or a remote control panel, which shall be capable of operating the RU. If a remote control panel is provided, it must be enclosed in a NEMA 4X watertight, metallic enclosure. The mode selector switch and temperature selection control shall be mounted in the control assembly. The control assembly shall be protected from mechanical injuries, environmental elements and mounted to the user can read, adjust and operate the TRCS and its controls when coupled to other TRCS units or other dry freight TriCon containers. The TRCS shall be designed to ensure display device is protected from UV radiation to prevent degradation of the visual readout. The controller panel shall allow fingertip control to:

Each touch key function must be permanently and clearly labeled. Relays, contactors, LED indicating lights and associated control devices shall be easily assessable to facilitate component troubleshooting and repair. With the exception of interior containers temperature set point, all user interface, troubleshooting and diagnostics shall not be performed through a logic keypad. If a micro process is provided, information shall be provided on the controller allowing the user to order the part. The information shall include, at a minimum, manufacturers name, address, phone number, part number, serial number, cost, etc…Spare fuses shall be over packed with each TRCS to support expedient field repair.

For Type II the following controls and indicators are also required:

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a) Battery Charge Level or Voltage Indicator
b) Fuel Level Indicator
c) Fuel level warning (for low fuel level)
d) Automatic Reset Switch/ Engine Protection
e) Battery fault indicator
f) Fuse diagnostic indicator (if fuses are provided)
g) High pressure refrigeration system fault indicator
h) Water/oil safety fault indicator
i) High/low temperature fault indicators for refrigerant circuits (both audio and visual with on/off switch)
j) Manual Preheat of engine driven refrigeration unit operational indicator
k) Operation of refrigeration unit operational indicator
l) Power saver On/Off of engine driven refrigeration unit operational indicator
m) Battery saver (prevents battery drain beyond the point that prevents restarting without re-charging the battery)

3.2.24 Ozone Depleting Chemicals. The refrigerant shall be commercially available, non-ozone depleting. Class I or class II ozone-depleting substances, as determined by the Environmental Protection Agency (EPA), shall not be contained in, and shall be prohibited from use during the fabrication of the TRCS system (see 6.6). Refrigeration unit shall be capable of operating without the refrigerant charge being maintained at a critical charge. The RU shall be equipped with a receiver and condenser that will allow the refrigerant to be "pumped down" and completely isolated in the receiver/condenser during maintenance and servicing operations as well as for transport and long-term storage. The liquid receiver shall not leak any refrigerant. The RU shall not leak more than .05% per year of the total rated refrigerant charge. The entire refrigerant system shall withstand pressures up to best commercial practices or the compressor manufacturers design limits without deformation of any components or any detectable leakage.

3.2.25 Refrigeration Service Valves. Two refrigeration service valves with standard ¼" male flare charging valves shall be provided. One for high side (discharge) and one for low side (suction). Caps shall be provided for all service valve fittings.

3.2.26 Power. The RU shall operate on worldwide voltages (208-230 VAC, 3-phase, 50/60hz power). All other replaceable components; i.e., relays, fan motors, solenoid valves, other control devices etc., (except the compressor) may operate from Direct Current (DC) or Alternating Current (AC). A manually re-settable circuit breaker or manually re-settable main "On/Off" switch shall be wired at the incoming power connection. All power connectors shall be heavy duty, water tight, corrosion resistant, strain relieved and protected from physical damage. The power draw of the refrigeration unit shall not exceed 3500 Watts in any operational scenario listed herein including the following rated containers ((at 0ºF and 35ºF set points at 100ºF and 125F (+0/-3°F) ambient)). For Type II, the refrigeration unit shall also operate seamlessly and properly from the engine driven RU.

The TRCS shall also be capable of being operated by power supplied from external Department of Defense (standard Army family of generators to include10k, 15k, 30k, 60k (MEP 806A&B models), 150k Mobile Electric Power and Tactically Quiet Generators) and commercial generators. Interface requirements per medium and large generator sets technical data.

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3.2.27 Power Supply Cables and Connectors. Option 1 (Army): Power input to the TRCS shall be from a MS90556C32412P, 60Amp, Class L, (MIL-DTL-22992), Connector Plug, Straight, attached to the free end of a suitable 12 foot SOO type cable. The fixed end of the power input cable shall be attached (hardwired) to a weather proof, fixed mount power input connection box, or directly to the refrigeration system unit. The plug shall be provided with the mating dust cap and lanyard chain. Cable mounting between the power input panel and refrigeration system unit shall be fixed to the container. The Power input point shall have a weatherproof circuit breaker installed between the power input connection and the refrigeration unit. A means of securing the power input cable during transport shall be provided.

For Type III (Air Force): the TRCS shall be supplied with a box mount, 45-degree, 60 Amp, Class L receptacle, MS90558-32412P, per MIL-C-22992 hard wired to a weather proof, fixed mount power input connection box, or directly to the refrigeration system. The receptacle shall be provided with the mating dust cap/plug and lanyard chain. The TRCS shall be fully compatible with the Air Forces Basic Expeditionary Airfield Resources (BEAR's) power system including: standard 60 Amp Power supply cables and connectors (NSN: 6150002558313), Power Distribution Panels and generators.

Note: Both options 1 and 2 shall be shall allow the RU to operate properly and safely using 208/230 volt, 3-phase, 50/60 Hz power and shall be suitable to handle the refrigeration unit's amp draw. The incoming voltage and current shall be stenciled on the connector or it's housing in legible black or white lettering. A high voltage placard/label shall be placed on the incoming power connector. The connectors wiring configuration shall be outlined in the documented wiring diagram and schematic. All power connections and cables shall be heavy duty, water tight, corrosion resistant, strain relieved and protected from physical damage. The entire power cabling and electrical system shall be rated to handle the current draw of the TRCS in all operational scenarios. All power supply cabling shall be safely stowed and contained within the ISO clearance allowance dimensions of the system.

3.2.28 Roof Load and access/steps/handholds. The roof shall be self-draining, prevent water pooling, and prevent water from both entering the container and/or causing component damage. The roof and its roof strength (per ISO 1496-1) shall allow less than ¼ inch permanent deflection. When the TRCS is tested in accordance with 1496-1, Weatherproofness test, and is level to within 5 degrees, the roof shall retain not more than 1/8 inch of water. The roof of the container shall support the weight of all personnel and equipment necessary to complete set up, tear down, operational procedures, rigging procedures and maintenance tasks which take place on the roof without evidence of damage or degradation. The roof shall be capable of supporting, without damage or degradation; (1) a uniformly distributed load of 40 lbs/ft^2 over the entire surface and (2) a concentrated load of 660 lbs over a 1 ft x 2 ft area, applied anywhere on the roof.

The TRCS shall have safe roof access steps/handholds enabling ground level personnel to safely access the roof level of the container. Each step/handhold shall be capable of supporting at least 400 pounds and, when stored for transport, shall not extend beyond the container ISO clearance allowances. All steps/handholds shall be permanently attached to the exterior of the container and shall comply with all transportation requirements contained herein. Applicable ISO-required safety label shall be supplied to warn of overhead danger.

3.2.29 Connecting Couplers, Quantity and Storage. Three (3) TRCS containers, dry freight TriCons or any combination shall be coupled together to form a series 1C equivalent 20' module container. The currently used couplers are NSN 3040-01-387-4048. Three couplers shall be supplied per TRCS

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container. The couplers shall be supplied in a fiber board box, labeled and placed inside the TRCS units.

3.2.30 Floor. The floor shall be watertight and shall be permanently attached to its bottom side rails and cross members. The floor shall be designed and installed to support the container's payload (P_g) as defined in ISO 1496-1, paragraph 6.1.1. The base of the container shall be constructed with provisions for 4-way insertion of fork tines using commercial and military material handling equipment. The floor shall be corrosion resistant, designed to maximize convective airflow and shall be non slip. The floor shall not be physically damaged during operational loading and unloading by warfighters. A flat or T-floor design is acceptable as long as the other requirements listed within this specification including NSF and internal pack out requirements are met.

3.3 Interface and Interoperability Requirements. The TRCS shall satisfy the following interface and interoperability requirements.

3.3.1 Dimensions. When three containers are coupled the dimensions, tolerances, and diagonal differences of the three containers shall meet the requirements for a 20-foot (1C) ISO container as specified in ISO 668 and 1496-1.

The external dimension shall not exceed:

Width	2,438 mm +0/-5	(8'-0" +0/-3/16)
Length	1,969 mm +0/-3	(6'-5 1/2 +0/-3/16)
Height	2,438 mm +0/-5	(8'-0" +0/-3/16)

The internal dimension (Nominal) shall be:

Width	2,137 mm	(7'-0 1/8")
Length	1,760 mm	(5'-11 5/8)
Height	2,126 mm	(6'-11 45/64")

Note: The internal dimensions of the RU shall not exceed those of the current Thermo King VM405 refrigeration unit in order to meet critical internal pack out requirements

The internal cubic volume shall be a minimum of:

7.56 cubic meters (270 cubic feet)

The minimum double door opening dimensions shall not be less than:

Width	1,758 mm	(5'- 9 7/32")
Height	2,089 mm	(6'-10 17/64")

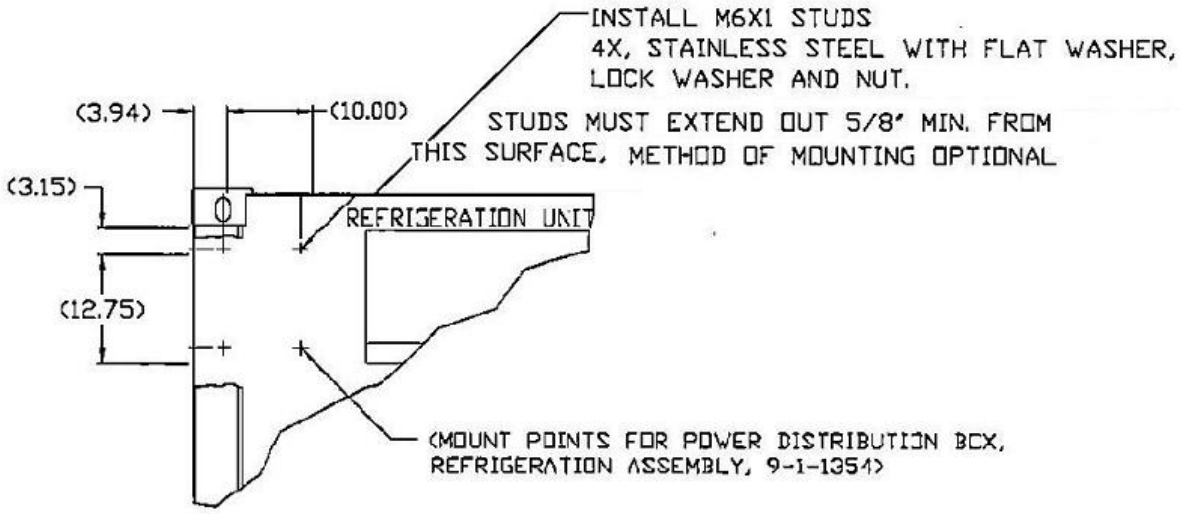
The minimum single man door opening dimensions shall not be less than:

Width	711 mm	(2'- 4")
Height	1905 mm	(6'- 3")

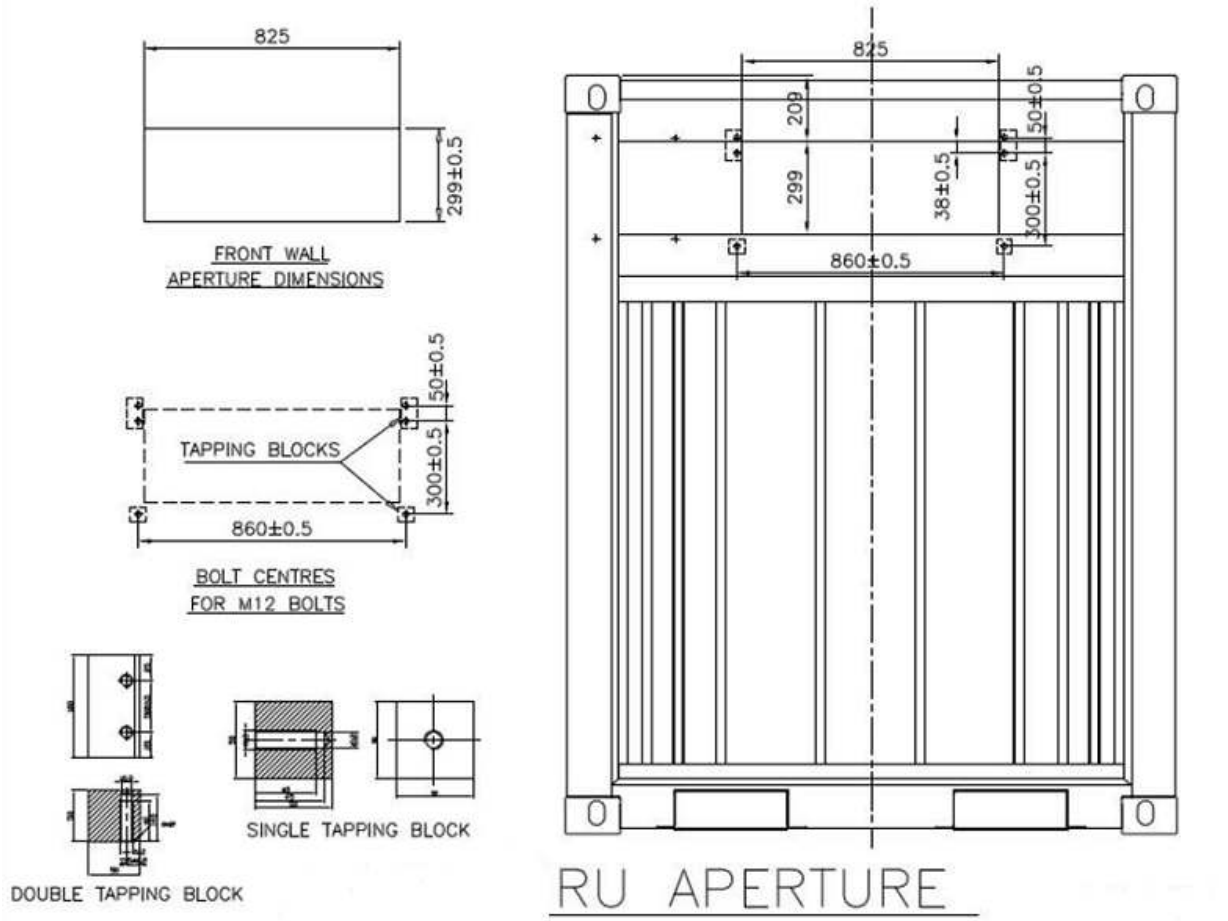
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3.3.2 Refrigerator Interface. The end wall opposite to the wall with doors (top of the road side) shall have an opening for mounting the refrigeration unit. It shall be located at such a distance from the roof and centered between the two walls and located in such a manner to maximize the internal cubic volume. The RU shall be located to promote easy troubleshooting, repair and replacement. The end wall shall use non-corrosive materials and be of sufficient strength to safely support the refrigeration unit having a minimum weight of 500 pounds (Type 1 and III) during all modes of military and commercial transport listed herein (1,000 pounds for Type II). The contractor shall ensure that the RU frame does not permanently deform when the mounting bolts/nut are tightened. The RU shall make a watertight seal against the rear wall of the container or any refrigerant or electrical lines passing through the wall shall be watertight. The TRCS container shall be outfitted with mounting points and hardware (See View A-A) to support the installation of the Power Distribution Box, Refrigeration Assembly, Natick Soldier Research Development and Engineering Drawing Number 9-1-1354.

Note: For Type I and III, for information purposes and compatibility with previously DoD developed mechanical refrigeration units, the TRCS shall be designed to accept/interface with the Thermo King VM405 MAX-EL (Model No. PBBD2) refrigeration units. The illustrations below shows the mounting and interface requirements.



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3.3.3 Padlock. Each container shall be supplied with padlocks conforming to ASTM F883 and shall be furnished with two keys per padlock.

3.3.4 Interchangeability and Ease of Maintenance Certification. The contractor shall certify and verify that: All component parts of the container, which are replaceable, are also interchangeable with the same parts of other TRCS furnished under the same model and contract and that all assemblies and installed attachments are accessible for preventive maintenance procedures; All assemblies and installed attachments are designed for ease of installation and replacement and are accessible for preventive maintenance actions and do not require the removal of other major assemblies and installed attachments.

3.3.5 Human Factors.

Human factors for labeling Standardization. Labels shall be located in a consistent manner throughout the equipment. The TRCS shall have permanent labels or placards indicating dangerous areas and components are present (i.e. “Danger, Rotating Fan, Hazardous Voltage, Lockout before Servicing, Warning Moving Parts, Sharp Objects, Heavy Objects, Caution Hot Surfaces” etc… All warning plates

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shall be properly placed for all potentially hazardous and hazardous locations.
Human performance and human factors engineering (HFE). The RU shall comply with all of the following human performance and HFE criteria in accordance with MIL-STD-1472 or DEF(STAN) 00-35, Human Factors for Designers of Systems, Part 17.
Human performance. The RU shall be operable by 5th through 95th percentile warfighters dressed in the Airman Battle Uniform (ABU) and Battle Dress Uniform (BDU) and cold-wet weather protective clothing, and protective ensembles for Mission Oriented Protective Posture. The RU shall also minimize personnel and training requirements within the limits of time, cost and performance trade-offs.
HFE. The RU design shall conform to Army accepted HFE criteria. The major aspects of those criteria relevant to RU are:
Simplicity of design. The RU shall represent the simplest design consistent with functional requirements and expected service conditions. It shall be capable of being operated, maintained and repaired in its operational environment by personnel with a minimum of training.
Hot surfaces. Any surfaces, which, in normal operation, expose personnel to temperatures in excess of 140°F as a result of inadvertent contact or 125°F during handling, shall be appropriately guarded. Surface temperatures induced by climatic environment and the condenser coil are exempt from this requirement.
Design for maintainer. The RU shall be designed for ease of maintenance. Standard parts shall be used whenever practicable. Mission critical items, which require rapid maintenance, shall be most accessible. When relative criticality is not a factor, items requiring most frequent access shall be most accessible. High failure rate items shall be accessible for replacement without moving non-failed items. Items shall be so located and mounted that access to them can be achieved without danger to personnel from electrical charge, heat, moving parts, chemical contamination, etc. Repairs to be made at the organizational level shall require no more than 4 hours to conduct. Repairs to be made at the intermediate level shall require no more than 12 hours to conduct.
The RU shall be maintainable using only standard mechanic and refrigeration tools specified. No special tools shall be used to perform maintenance or installation/removal of the RU.
Error proof design. Provision to assist assembly and installation, and preclude improper assembly or installation, shall include the following: a. Physical measures to preclude interchange of items of a same or similar form that are not in fact functionally interchangeable. b. Physical measures to preclude improper mounting of units or components. c. Measures (e.g., coding) to facilitate identification and interchange of interchangeable items. Measures (e.g., alignment pins) to facilitate proper mounting of items. d. Measures to insure that identification, orientation, and alignment provisions include cables and connectors. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more parts possessing such functional and physical characteristics as to be equivalent in performance and durability and capable of being exchanged one for the other without alteration of the parts themselves or adjoining parts, except for adjustment, and without selection for fit or performance.

Relevant anthropometrics measurements are listed in Table II, and additional guidance can be found in the 19 Anthropometrics Survey of U.S. Army Personnel (see 6.10). MIL-STD-1472 shall be utilized as

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an additional guide. All HFE concerns must be properly addressed and recommended procedures incorporated into the technical manual.

TABLE II. Anthropometrics dimensions.

Description	Dimension to be used	Example application
Overhead reach	73 inches (5th percentile female)	Shelf height
Functional reach	32 inches (5th percentile female)	Counter depth
Stature	74 inches (95th percentile male, boots)	Head clearance
Shoulder breadth	27 inches (95th percentile male, bulky clothing)	Passage clearance

3.3.6. Test of the performance of a container with refrigeration by a mechanical refrigeration unit (MRU). The TRCS shall undergo and pass ISO 1496-2, Test 15a.

3.4. Support or Ownership Requirements. The TRCS shall meet the following support or ownership requirements.

3.4.1 Weight. The TRCS shall not exceed the following weight ratings listed below. TRCS loading configurations shall not exceed the load handling capabilities of the vehicle that is transporting the TRCS and shall meet all of the weight restrictions dictated by the ISO certifications. The weight of the refrigeration unit shall not exceed 500 lbs. The refrigeration unit can be a one-piece unit or a two-piece unit. Lifting provisions and/or forklift pockets shall be provided for safely and securely mounting the Refrigeration Unit on the container.

The TRCS shall meet the following Weights and Ratings for a single container:

Maximum Gross Weight (R1)	11,090 kg	(24,450lbs)
Tare Weight (Max. including the refrigeration unit) (T1)	2,000 kg	(4,410 lbs)
Payload (Min.) (P1)	9,089 kg	(20,040 lbs)

The TRCS shall meet the following Weights and Ratings when three containers are connected.

Maximum Gross Weight (R1)	24,000 kg	(52,910 lbs)
Tare Weight (Max.) (T1)	6,000 kg	(13,228 lbs)
Payload (Min.) (P1)	18,000 kg	(39,682 lbs)

Note: The TRCS shall meet the ISO 9 High Stacking requirement as a 1C container at 24,000 kg Max. Gross Weight

3.4.2 Structural Requirements. The TRCS shall be structurally suitable for inter-modal conveyance by air, road, rail and sea, including interchange between these forms of transport.

3.4.3 Corner Fittings. The TRCS shall have bottom corner fittings located on all eight corners to accept the specified interconnectors. The corner fittings shall conform to the dimensional and strength requirements of ISO 1161. The corner fittings shall be designed to facilitate handling, stacking and

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restraints as well as coupling to form two and three array configurations for efficient handling and transport.

3.4.4 Forklift Pockets. The TRCS container shall have forklift pockets for four (4) way forklift entry/access and be correctly spaced, sized and positioned to allow the TRCS to be safely lifted and moved (both empty and fully loaded) in the field using military and commercial fork lifting equipment (with proper lifting capacity) without causing any damage or deformation to the TRCS, it's cargo or the forklift. The height, width, and spacing dimension of each fork pocket shall conform to ISO 1496-1, Annex C. At a minimum, fork pocket opening shall be 12.0 inches wide by 4.0 inches high. The forklift tube opening and the walls shall be reinforced to prevent any physical damage to the container.

3.4.5 Interior and exterior lighting. For Type III, the TRCS interior shall have a vapor and shock proof lighting fixture or fixtures mounted on the interior of the IC. The fixture(s) shall have a protective cover, which is shatterproof and removable without special tools for bulb replacement. All wiring shall be shielded from physical damage in electrical conduit and conform to NEC standards. All conduit and lighting components shall be located to prevent contact with other interior components and be easily accessible for maintenance, repair or servicing. All conduit and lighting components shall be designed for long life and withstand extreme cold, heat and thermal shock without degradation. The lighting system shall be designed to withstand all of the operational, environmental and transportability requirements listed herein without failure. The fixture shall accept and be supplied with a minimum 100 watt, 120 VAC incandescent light bulb(s) or LED lighting system with equivalent lumens. The switch for the light shall be located inside and adjacent to the door and shall be vapor tight and protected from damage. The fixture shall be installed in a location to prevent light obstruction. The interior lighting system shall be designed and operate with full blackout mode capabilities. The interior light shall be designed to plug into the weather tight exterior grade 15 Amp, GFCI duplex outlet mounted on the exterior of the container.

A heavy duty portable exterior light system (PELS) with a mounting bracket shall be supplied with and over packed with each TRCS (Type III). The PELS shall attach and be removable from the exterior of the container without special tools and shall provide focused light on the door side of the container to support safe access/egress/loading and unloading by personnel in no or low light conditions. The PELS shall be supplied with a minimum 25 foot, 15 Amp, 3 prong 120VAC power cord that interfaces with the weather tight GFCI duplex outlet on the exterior of the container or can be powered from alternate 120 VAC, 60 Hertz shore power. The mounting bracket for the PELS shall be supplied with an integrated weatherproof/watertight light switch that is located near the door entrance and less than 6 feet high from the ground. The PELS shall be designed to withstand the rigors of repeated field use without degradation or damage. Each PELS shall be supplied with one spare bulb. When deployed, the PELS shall be subjected to and pass all of the operational tests listed herein. The PELS shall be by subjected and pass all of the operational tests listed herein. When stowed, the PELS shall be subjected to and pass all of the structural and transportability tests listed herein.

3.4.6 Multi Fuel Engine Driven Refrigeration System (Type II only). The refrigeration system shall be a multi-fueled (JP-8 and diesel) engine driven system. The engine driven section shall be an integral part of the refrigeration unit and provided with a drain valve to drain the fuel tank in the integral engine-driven system. The system shall have the capacity to generate sufficient power to start and continuously operate the refrigeration system. The refrigeration system must be able to operate properly from the integral engine while stationary, and while

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being transported by truck, rail or externally airlifted. For Type II, the TRCS shall be provided with switching device capable of changing from integrated engine driven to an external electrical power source.

The engine selected shall have the following features unless air cooled:

1. Coolant level gauge
2. 50/50 anti-freeze
3. Radiator coolant reserve
4. Heavy-duty coolant hoses.
5. Engine coolant high temperature cutout.
6. Engine low oil pressure cut-out
7. Power switch with multiple positions for each of the power sources referenced and an OFF
8. Main Power Switch.
9. Battery
10. Auxiliary Fuel Tank

Note: air cooled engines are acceptable if they meet the operational and environmental and transportability requirements listed herein.

The engine driven RU control panel shall have the following features:

a. Battery Charge Level Indicator
b. Fuel level indicator
c. Fuel level warning (for low fuel level) Note: Can be located on the fuel tank

The TRCS shall be provided with an auxiliary fuel tank to store and provide fuel for the refrigeration unit when operating in engine driven mode. It shall be DOT certified and shall provide enough fuel to keep the refrigeration operating at 100% capacity, during the worst ambient conditions specified herein for a minimum of twenty-four (24) hours without refueling. The tank interior surface shall be permanently protected from corrosion. The tank shall include a protected fuel gage that is clearly visible to the user. The tank shall also include; baffle plates as required, essential pressure relief valves, filler cap, drain valve and a captive retained inlet fuel screen. The drain valve shall allow the tank to be completely drained prior to transport. These devices shall prevent fuel spillage when the container is operated up to a maximum of a 20 percent slope/grade in all axis. A highly durable, abrasion resistant fuel line shall be provided which transfers fuel between the auxiliary fuel tank and the engine driven RU. A captive retained cap for the fuel line fitting shall be provided. The auxiliary fuel tank shall be attached to the container frame in a manner to facilitate easy removal and shall be properly grounded. Provisions shall be made for storage of any excess length of multi-fuel engine driven RU auxiliary fuel line near the auxiliary fuel tank. The auxiliary fuel line shall be installed/located in such a way so as to minimize and eliminate operational and safety hazards. Electrical and fuel lines shall not be run adjacent to one another. These hazards include, but are not limited to, electrical, mechanical vibrations, abrasion, puncturing, and environmental issues such as exposure to UV radiation. The type and quantity of fuel shall be permanently stenciled onto the fuel tank.

3.4.7 Data Plates and Marking. Each TRCS container shall have corrosive resistant metal identification plates permanently affixed to the container. The data plates shall contain the following information:

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TRICON REFRIGERATED CONTAINER SYSTEM (TRCS)
NSN: (see 6.2)
LIN:
Serial Number:
Model No:
Tare Weight, Equipped: LB_____ KG_____
Payload Capacity: LB_____ KG_____
Gross Container Weight:
Contract Number:
Manufacturer Name, Address and Telephone Number
Manufacturers Part Number.
Date of Manufacture: (Month & Year)
Technical Manual No:
Allowable Stacking Weight @1.8G: _____lbs
Racking Test Load Value: _____lbs
(Note: Tare weight. The tare weight marked on the container shall be the weight of the container, plus the weight of the refrigeration unit).

TRCS REFRIGERATION UNIT:
NSN:
Weight:
Refrigerant Type and Quantity:
Power: (Volts, Amps, Nominal Current):
In Rush Current:
Manufacturer Name, Address, Telephone Number
Manufacturers Part Number.
Date of Manufacture: (Month & Year)
Technical Manual Number:

Note: for Type II include engine performance data, manufacturer, horsepower, oil quantity and type, coolant quantity and type, etc…

3.4.8 Sanitary Requirements. All surfaces inside the TRCS shall be smooth, sealed, and capable of being; sanitized, cleaned by steam, and high pressure hot water washer with detergents. The TRCS shall be designed to the criteria of Standards No. 2, C-3. The container interior and exterior walls, ceilings, floors shall not degrade or deteriorate after cleaning with a high-pressure (90 PSI) hot water system. Critical system parts (i.e., condenser/evaporator) shall be accessible for cleaning/disinfecting without the removal of the RU from the container to prevent the build-up/growth of germs/viruses. No pockets/voids/joints etc. shall exist in the system that cannot be reached or cleaned via conventional cleaning methods. The interior surface of the container shall facilitate cleaning by soap water and pressure washer without damage or detriment of operation. The floor shall be flat and/or slightly pitched towards the floor drain(s). The TRCS shall be NSF Standard No. 7 certified and an up to date listing shall be provided to the Government and kept on file for future Government inspections and the NSF listing mark shall be affixed on the exterior of the TRCS (on or near the data plate).

3.4.9 Exterior Markings. The contractor shall obtain the TRCS container serial numbers and any optional customer required markings from the procuring agency. All markings shall be permanent and remain clearly visible/legible throughout the entire life of the TRCS. Each TRCS shall be marked in

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accordance with ISO 6346 and example in Figure 1 black paint with Green TRCS or black paint with Desert Tan TRCS in accordance with ISO 6346 with the following additional requirements:

a. "CENTER OF BALANCE, EMPTY" shall be stenciled at the longitudinal center of balance of the empty TRCS. The container shall be stenciled on both sides with a 4-inch high vertical arrow and lettering 2 inches high at the longitudinal center of balance.

b. "TECHNICAL MANUAL HOLDER" shall be stenciled on the manual holder in black lettering.

c. The following lettering with applied appropriate weights and data shall be stenciled on the exterior door of the TRCS.

MAX.GROSS WEIGHT	Kg and lbs
TARE WEIGHT	Kg and lbs
MAX PAYLOAD	Kg and lbs
CUBIC CAPACITY	M^3/CU.FT.

The containers "vertical height" shall be stenciled in both meters and feet with 4-inch minimum height lettering on the exterior of the container.

Any other safety markings, required for warning personnel of potential safety hazards that may be encountered upon removal/relocation of components, guards and/or doors shall be provided.

3.4.10 Code Markings. The manufacturers production number shall be stamped on the bottom left door end casting and the military serial number shall be stamped (minimum ½ inch high), on the bottom right door end corner casting.

3.4.11 Approval Plates and Consolidated Data Plate. The TRCS shall be Design, Tested and CSC Approved in accordance with US Coast Guard regulations by one of the following major classification societies:

- Lloyd's Register North America, Inc (LRNA)
- A.B.S. American Bureau of Shipping (U.S.A.)
- B.V. Bureau Veritas (France)
- G.L. Germanischer Lloyd (Germany)

The TRCS shall be tested and classified as an individual unit and when three units are connected to form an ISO Series 1C 20' module configuration at 52,910 pounds/24,000 kilograms using US Military approved interconnectors. Up to date approval numbers, certificates, and test reports shall be provided to the Government during First Article Testing and kept on file and made available to the Government for inspection. The plate, plates or plaques shall be displayed as required by the convention, in accordance with CFR 49, Chapter 4, parts 450, 451, 452, and 453. Any additional requirements of the approval authority shall be met. Each container shall be affixed with the seal of the approval authority. The TRCS Consolidated Data Plate or equivalent shall meet the requirements of 49CFR, subpart C.

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3.4.12 Instructions and Warning Placards. The TRCS shall have appropriate warning labels and instructions to assist personnel with safe operation and emergency procedures. Conspicuous labels or placards shall be placed on, or adjacent to, any equipment that presents a hazard to personnel (e.g., high voltage, heat, toxic vapors, explosion, or radiation). These labels or placards shall describe the hazard and state appropriate precautions. Labels and placards should also describe the consequences of not complying with the stated warning. All placards shall be permanently affixed to equipment, and shall be sufficiently durable to remain legible at all times throughout the entire systems life. All placards shall be easily noticeable, recognizable, and understandable. Permanent metal wiring diagrams and electrical schematics shall be mounted to the RU or TRCS in a location accessible and readable to the service technician.

3.4.13 Refrigeration Unit Identification Data Plate. The contractor shall supply the refrigeration unit with an identification data plate and placards permanently affixed to its outer surface that is easily readable and shall be sufficiently durable to remain legible at all times.

3.4.14 Reliability. The TRCS shall demonstrate a Mean Time Between Hardware Essential Function Failure (MTBHEFF) of not less than 1600 hours at the 80% lower confidence limit. Throughout this period, the TRCS shall operate properly and maintain internal set point temperature conditions at thermostat settings as specified in Section 4. Note: The Government intends to procure three FAT units to support RAM testing.

3.4.14.1 Service and Maintenance. All major components of the TRCS including the RU (i.e. expansion valve/metering device, fans, heaters, thermistors, sight glass, sensors, filter dryer, etc.) shall be accessible for adjustment, maintenance, servicing, and replacement without removing any other major component with the exception of the heat exchanger coils and compressor. All other components and assemblies shall be designed for ease of installation and replacement and shall be accessible for maintenance, service, and replacement with minimum disturbance to other components. No special tools shall be needed for any maintenance, repair, service, installation or replacement action. Except for Type II, all motors and compressors shall be direct drive to negate the need of using belts and refrigerant leakage through a shaft seal. All relays and contactors shall be removable without disrupting other systems and subsystems, not controlled by that relay or contactor, from operating. Condenser and evaporator fins shall be protected from any damage that might occur during operation, service, and maintenance actions. Evaporator housing will have a NSF metal screen at the air discharge point to prevent items from being stored in the evaporator and condenser section and preventing hand/figure injury from sharp coils fins. When removing the condenser cover to gain access to major components (i.e., compressor, electrical control panel, filter dryer, refrigerant sight glass, fan motors, valves and solenoids) the system shall not negatively affect system operating capacity, head pressure or hinder operation of the unit in anyway. The RU shall require only annual maintenance when in storage. If a semi hermetic compressor is provided, maintenance personnel shall be able to visually check the moisture content of the refrigeration system and oil levels. If a fully hermetic compressor is provide, an oil sight glass is not required. All points requiring lubrication shall be equipped with standard fittings and shall be accessible from the outside of the unit. Each unit shall have the fittings or location tagged to indicate the temperature and grade of lubricant used and recommended frequency of lubrication. The RU shall be equipped with lifting provisions to facilitate safe installation and removal from the IC. RU shall be designed so that it will rest on the ground without the aid of any support brackets, forklift or hoist during maintenance and inspections. Cleaning coils, adjusting/replacing expansion valves, thermistor, resistive heaters, electrical connectors and temperature sensors shall be accomplished without requiring the removal of the refrigeration unit from the IC. The RU shall be safely removable

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by using either a forklift or crane/pulley assembly. Eyelift provisions and/or forklift forks shall be installed to allow safe forklift and/or crane lift. The median time to repair (MedTTR) for all essential unscheduled maintenance demands (EUMD) must not exceed 1 hour. The maximum time to repair (MaxTTR) for 90 percent of all EUMD must not exceed 1.5 hours. The maintenance ratio (MR) must not exceed .01 maintenance man-hours/operating hours.

3.4.15 Convention for Safe Containers (CSC) Certification. The TRCS container design and two additional connected TRCS shall be certified in accordance with 49 CFR, Parts 450 through 453. The rated stacking capability shall be for three (3) TRCS's coupled together as a 1C container supporting loads in accordance with ISO 1496-1.

3.4.16 Electrical System Wiring and Refrigeration Component Certifications. The RU shall be designed to comply with the National Electrical Code (NFPA 70) and as specified herein. Cables and wires shall be secured to ensure that cables or wires do not hinder or obstruct equipment maintenance, and to prevent chaffing from contact with adjacent structures. Cables and wire harness runs shall eliminate or minimize the use of cable splicing. Electrical isolators shall be used between all wiring and metal components to prevent abrasive action to wire insulation.

Refrigerant containing components used in the refrigeration units shall be certified as conforming to UL Standard 207 and limit controls used shall be certified as conforming to UL Standard 353. Verification of certification shall be conducted via submission of the manufacturer's certification, a certified test report, appropriate labeling or listing marks (by submission of the manufacturer's certification), a certified test report, or other appropriate labeling/listing marks.

3.4.17 Wire marking. All wiring shall be marked to indicate points of origin and ending, using nomenclature, which is consistent with the electrical schematics, and wiring diagrams supplied with the RU. (i.e. color code, numbering etc).

3.4.18 Transportation. Individual TRCS containers shall be able to be coupled together or with other dry freight TriCons to form an ISO 1C (8'x8'x20') equivalent. These configurations shall be transportable by air, highway, marine and rail assets as specified herein. The containers shall be capable of being transported by military and commercial vehicles, and trailers, trains, marine vessels, aircraft, dolly sets and shall withstand the impact forces encountered in shipment without damage or permanent deformation. The TRCS shall be sufficiently rugged to withstand field operation, maintenance, and transport within the environmental conditions specified herein. When exposed to these conditions, the TRCS shall show no evidence of structural damage, misalignment or malfunction of components, leaks, fractures, abnormal vibration, or any other irregular operation.

3.4.19 Internal Air Transportability. The TRCS shall be transportable via C-130, C-17, and C-5 and other Air Mobility Command (AMC) aircraft. The TRCS shall comply with all applicable safety, loadability, and tie down restraint guidelines. The TRCS container shall incorporate an operable hatch or pressure equalization device to preclude damage that would endanger the aircraft or personnel due to pressure differential which may be encountered during rapid in-flight decompression of as much as 8.3 psi within 1/2 second or less per MIL-HDBK-1791. Additionally, MIL-STD-209 and MIL-STD-1366 shall be used as a guide.

For reference, the following shall be used as a minimum for structural design load factors under full payload conditions.

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Forward:	3.0g
Aft:	1.5g
Lateral:	1.5g
Up:	2.0g
Down	4.5g

- Loads to be applied individually
- Force of gravity is compensated for in values

The margin of safety, based on the ultimate strength, shall be at least 1.5 factor.

For Type II only, a five gallon capacity polyethylene (Jerry) fuel can, complete with mounting hardware and quick connect fuel hoses to the RU, which is required to comply with internal air transportability requirement as the TRCS cannot be transported on Military aircraft with fuel in the system.

3.4.20 External Air Transport. The TRCS shall be capable of being externally transported (sling loadable) by a DoD rotary wing aircraft individually and in the coupled state to include, but not limited to CH-47D helicopters. The TRCS in a single, dual and triple (1C) configuration, shall be transportable for a minimum mission for 30 nautical miles at a maximum weight determined by the test agency based on an elevation of 2,000 feet at an ambient temperature of 70ºF. The TRCS shall be structurally capable, inspected and tested for conformance in meeting the requirements of interface standard MIL-STD-209 and shall show no signs of damage or degradation as a result of aerial transportation. Tie down provisions of MIL-STD 209, design criteria of MIL-STD-913 and helicopter lift limits of MIL-STD-1366 shall be used. This includes static lift test, proof load test and actual flight testing.

3.4.21 Rail Transportability. The TRCS shall be rail transportable in CONUS and NATO countries without restrictions in the 1C configuration. The TRCS shall be rail impact tested with three TRCS coupled together or a combination of TRCS and dry freight TriCons as a 1C container in accordance with MIL-STD-810, Method 516.5, Procedure VII (rail impact) with the TRCS at its maximum gross weight. The approval shall be supplied by the American Association of Railways (AAR) or the Surface Distribution and Deployment Command, Test Engineering Agency (SDDC-TEA).

3.4.22 Ground Mobility. The TRCS (coupled and non-coupled), when mounted on a prime mover (Pallateized Load Handling System (PLS9 or PLS trailer, shall be capable of transportation over primary roads, secondary roads, and cross-country terrain. When coupled to other TriCons or TRCS or a combination of both, the TRCS shall be capable of unrestricted highway transport and standard military flat rack systems used. The TRCS shall be capable of operating properly and maintaining set point temperatures after each ground mobility test. For Type II, the RU shall operate from the integrated multi fuel engine. The TRCS shall withstand the shocks and vibrations encountered in ground transportation and operate properly without damage or degradation to the TRCS. The TRCS shall have ground mobility characteristics to withstand shock and vibration encountered during military operations. The empty and fully loaded TRCS, when mounted on the tactical vehicle or trailer, shall show no permanent deformation and operate properly after being subject to the following tests: Perryman Road Test, Munson Gravel Road Test. The TRCS shall meet the requirements of MIL-STD-810 Test Method 516.5, Procedure VII. Note that the empty configuration, for each prime mover refers to the TARE weight of the TRCS. The fully loaded configuration, for each prime mover, refers to the maximum allowable weight that can be lifted and/or transported by prime movers or the Max. Gross

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Weight of the TRCS. Transit Drop (transport mode): A TRCS shall be couple to two other dry freight TriCons in the outboard position and subjected to and pass the 6-in. transit drop test in accordance with MIL-STD-810G, Method 516.6, Procedure IV-Transit Drop.

3.5 Environmental Requirements. The TRCS shall operate under the following environmental conditions, without degradation in performance. The container must be capable of uncovered long term storage in extreme hot and cold climate, high humidity, blowing sand, dust, heavy snows, mud, soft ground, earthquakes, debris, or combination of the above, as defined in AR 70-38 to include stowage on PREPO afloat vessels up to 30 months without loss of mission essential functions with routine unit-level (i.e., -10 level) maintenance performed every six months.

3.5.1 Ambient Conditions. The TRCS shall set up, tear down, operate (maintain set point temperatures) and function properly, in all ambient conditions of both hot and basic climatic categories as defined in AR 70-38. These are to include subcategories of Hot-Dry, Hot-Humid, Constant High Humidity, Variable High Humidity, Basic Hot and Basic Cold Environments. This is to account for all contributing factors of ambient air temperatures, solar radiation, ambient relative humidity's, induced air temperatures and respective induced relative humidity. Also, it is to account for each of their respective combinations as outlined in the document. The worst-case scenario for cooling capacity, unless otherwise explicitly stated, refers to operational conditions of Hot-Dry Climatic Category A1 with 125°F (+0/-3°F) ambient with 3% relative humidity. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.2 Weatherproofness. The TRCS shall not exhibit water leakage when subjected to the ISO 1496-1/2 weatherproofness test. The TRCS shall not permit water intrusion into the container, or cause operational problems with the refrigeration unit. Upon completion of testing, verify that no water has leaked and/or penetrated into the container, electrical control panels, and the refrigeration/heating unit shall function properly.

3.5.3 Salt Fog. The exterior of the TRCS shall be capable of withstanding a 5% ± 1% concentration of salt fog for extended periods as will be encountered during transportation by or storage on sea vessels. The TRCS shall operate properly when exposed to a marine salt fog environment and the Salt Fog Test of Method MIL-STD-810 Method 509.4 without experiencing corrosion, or other forms of degradations except for light corrosion in the normal wear locations including fork lift pockets and ISO corner fittings.

3.5.4 Humidity. The containers shall not be affected by operation or storage in a high humidity natural environment. Temperature ranges of 80°F to 95°F and relative humidity levels of 74% to 100% shall have no affect on the performance of the system. The TRCS shall be subjected to and pass the Humidity Test Method 507.4 MIL-STD-810, without evidence of corrosion, swelling, structural deterioration, degradation, operational failures, property alterations or permanent deformation. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.5 Blowing Sand and Dust (operational mode). The TRCS shall remain operable and shall not permit infiltration of sand/dust into the food storage area and evaporator section and shall sustain no damage or degradation in performance in sand and dust conditions that cannot be corrected by routine maintenance by the TRCS operator. The TRCS shall be subjected to and pass MIL-STD-810, Test Method 510.4, Procedures I and II. For Type II, the RU shall operate from the integrated multi fuel engine.

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3.5.6 Fungus. The TRCS shall be impervious to fungus growth. No TRCS component shall contain any type of fungus sustaining nutrients. All TRCS components shall successfully pass the Fungus test Method 508.5 of MIL-STD-810 or by analysis and raw material supplier certifications.

3.5.7 Transportation Operations. Transportation of the TRCS shall not be affected/ hindered by any type of environmental conditions and shall operate properly in all potential scenarios. This is to include, but not limited to, humidity, rain, freezing rain, snow, ice, sand, dust, elevation, high ambient temperatures, solar loadings, and in low ambient temperatures (down to -35°F) and other parameters as required by 3.5.1.

3.5.8 Storage and Transit Temperature. The TRCS shall not be damaged by storage or transit under the conditions specified in Table I. The TRCS shall set up, tear down, operate and function properly following storage, in all ambient conditions of both hot and basic climatic categories as defined in AR 70-38. These are to include subcategories of Hot-Dry, Hot-Humid, Constant High Humidity, Variable High Humidity, Basic Hot and Basic Cold Environments. This is to account for all contributing storage and transit induced air temperatures and humidity's. Also, it is to account for their respective combinations as outlined in the document. Certification shall be provided by the contractor. For Type II, the RU shall operate from the integrated multi fuel engine.

Table I: Storage and Transit Temperatures

Air Temperature	
Degrees F	Degrees C
+ 160 - 40	+ 71.1 - 40

3.5.9 Noise. The noise levels produced by the TRCS shall comply with MIL-STD-1474D, Noise Limits. The noise at the operators and servicing positions shall not damage personnel hearing while operating. The noise level shall not exceed 85dB. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.10 Solar Radiation (Sunshine). The TRCS shall show no evidence of structural damage, degradation, or permanent deformation as a result of exposure to ultraviolet effects. The TRCS shall operate as intended and maintain proper operating conditions throughout the entire test duration of Method 505.4 (Solar Radiation (Sunshine)) of MIL-STD 810. Testing will be conducted on a TRCS with the color as specified in the contract. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.11 Finishing, Cleaning, Treating, and Painting. The containers shall be made of composite, aluminum or steel material, inherently corrosion resistant, and treated to provide corrosion resistance for an expected life of 20 years. The containers shall be cleaned, pretreated, primed and painted in accordance with the coating manufacturer's recommendations for the specific coating system that will be used. With the exception of muffler grade stainless steel, all steel surfaces shall be treated with zinc rich primer. Coating system shall be selected to provide resistance to environmental degradation and paint chipping. The interior surfaces of the container and the RU bulkhead shall be food grade, corrosion resistant, NSF approved stainless steel. Any welds and heat affected zones in galvanized steel shall be coated by zinc rich primer or thermal spray zinc to restore the zinc finish or provide

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equivalent protection provided by the zinc. Galvanized steel shall be pretreated with zinc phosphate. Surfaces not painted shall be treated against corrosion or shall be fabricated of materials that are inherently corrosion resistant. The paint shall not show signs of chips, scratches, orange peel, discoloration, sag, wrinkle, smudges or lifting. The finish shall be smooth and free of dirt or contaminants. No deterioration, corrosion, or damage to materials or material finishes shall be observed as a result of exposure to the environmental conditions specified herein.

The any coating the following minimum standards shall be met. The top coating thickness shall be between 2.0-3.5 mils and not less than 2.0 mils thick. The impact resistance shall meet ASTM D2794 at a min of 120 in/lbs. The adhesion shall meet D3359 with zero detachment. The flexibility per ASTM B522: PASS 1/8" MANDREL. The minimum melting point of 194°F, Maximum 1/8" Creep after 1000 HRS. The hardness per ASTM D3363: 2H-3H.

The topcoat color shall be determined in the solicitation or purchase contract per DESERT TAN color chip 33446 or Camouflage Green, Color Chip 34094 per FED-STD-595 unless otherwise specified.

The container materials shall conform to the minimum following requirements:

- Metal Preparation
  - All steel prior to forming - shot blasted to Swedish Std So.2-1/2 (SSPC-SP10) followed by application of shop primer
  - After forming and assembly – all weld areas shall be shot blasted and primed
  - All MGSS panels degreased and sweep blasted with non-metallic media

- Steel frame and under frame
  - Shop primer zinc rich – 10 µm
  - Second primer zinc rich – 20 µm
  - Intermediate primer polyamide epoxy – 40 µm
  - Top coat polyurethane – 50 µm
  - Total dry film thickness - 120 µm min

- MGSS Panels
  - Primer polyamide epoxy – 50 µm
  - Top coat polyurethane – 60 µm
  - Total dry film thickness – 110 µm min

3.5.12 Ice and Freezing Rain. The TRCS shall be designed such that its performance is not degraded or affected by ice, snow, rain and/or freezing rain. The TRCS shall not allow the accumulation of water in such a way as to cause failure of operation or damage upon freezing. The TRCS shall be designed such that ice/freezing rain does not bind moving parts, add significant weight, increase hazards/failures, or reduce efficiency. The TRCS shall meet the requirements for ice and freezing rain test per MIL-STD-810, Method 521.2. Failure to meet the requirements shall constitute failure. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.13 High Temperature. The TRCS shall meet the requirements listed in the high temperature test as outlined in MIL-STD-810, Method 501.4 for the Design Type climatic designation of hot. TRCS component parts shall not experience damage, be deformed in any way or fail to operate properly

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following exposure to both the Storage and Operational Procedures defined in Method 501.4. This is to include, but not limited to, sealing properties, control circuits, required clearances, power supplies, system capacities, discoloration, melting, dimensional changes, lubrication systems and other related effects. The RU shall start and operate properly in ambient temperatures up to 125°F (+0/-3°F). For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.14 Low Temperature. The TRCS shall start, operate and maintain set point temperatures in a Basic Cold environment as defined in AR 70-38 and shall meet the requirements listed in the low temperature test as outlined in MIL-STD-810, Method 502.4 Procedure I (Storage) and Procedure II (Operation). For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.15 Blowing Rain (operational mode). The TRCS shall operate in rainfall of not less than 4 inches per hour at 40 mph per MIL-STD-810F, Test Method 506.4, Procedures I and II. For Type II, the RU shall operate from the integrated multi fuel engine.

3.5.16 Electromagnetic compatibility. The TRCS shall not exceed radiated emission (RE) 102 limits for ground applications (Navy fixed and AF) over a range of 2 mega Hertz (MHz) to 400 MHz or conducted emission (CE) 102 limits of MIL-STD-461.

3.5.17 Shelving (Option specified on contract). The TRCS shall be provided with an optional NSF approved (Food Grade Stainless Steel), removable shelving system consisting of five (5) shelves per side, ten (10) shelves total. Each shelf shall be a minimum of 22” deep and the entire interior width of the container. The top shelf can be notched to clear the evaporator housing, if necessary. Each shelf shall be capable of supporting a 600 pound (distributed load) per shelf while stationary without any structural damage to the shelving, the container or the load. The shelving system shall also be capable of holding six (6), three (3) per side 95% percentile human cadavers (one per shelf) in body bags without contacting one another or causing physical damage to the shelving system, the container or cadavers. Shelving shall be removable to facilitate cleaning and shall be capable of being positioned on either wall of the aisle way. The unloaded shelving system shall be capable of withstanding the all of the transportability requirements listed herein without physical damage to the shelving, or the interior of the container. Shelving shall be secured in the TRCS interior to withstand all transport and dynamic requirements without damage to the shelving, the cargo or the TRCS. Shelving shall be installed so that shelves do not cover the IC floor drain opening or hinder IC box drainage and maintainability of the floor drain. Shelving shall not hinder accessibility/maintainability of the evaporator section. The shelving system assembly and disassembly shall not require any special tools and be designed for easy access to mounting hardware. The shelving system shall also be designed for the collection and pooling of fluids without spilling on the floor or the lower shelves and designed to store Unitized Group Ration boxes without wasted space. Note: Unitized Group Ration boxes are 24” long X 14” inches wide X 9.5” inches high. Each shelf shall be supplied with captive tie down straps or a netting system to prevent the movement and/or damage of the cargo and the TRCS. The shelving system shall have no loose parts when assembled inside the TRCS container.

4. VERIFICATION

These stated verifications are recommended, but alternatives may be proposed by the contractor and must be approved by the appropriate Government entity prior to commencement of testing.

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4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1)
b. Conformance inspection (see 4.1.2)

4.1.1 First Article Inspection. When a first article inspection is required, it shall be performed on an individual TRCS as well as other TRCS/dry freight TriCon configurations connected together. For the individual TRCS, the inspection shall include the in depth examination per Table II and the verifications listed in Section 4. The single (1) and three (3) TRCS/TriCon assemblies connected together shall be tested in accordance with the following paragraph. Nonconformance to any specified requirement or the presence of one or more defects shall be cause for rejection. One TRCS shall be provided to the procuring agency or designated US locations (possibly two) to support Government sponsored pack out verifications.

Using the coupling connectors, three TRCSs shall be connected to form an integral 20-foot module. The 20-foot module shall then be subjected to all the tests as described in ISO 1496-1. The weatherproofness test shall be performed as the last ISO 1496-1 test. In addition, the coupled TRCS shall be certified for CSC 9-high stacking, tested in accordance with ISO 1496-1. Nonconformance to any specified requirement, the failure of any test, or the presence of one or more defects shall be cause for rejection.

4.1.2 Conformance Inspection. Conformance inspection shall consist of all the verifications specified in Table II.

TABLE II. Verification Methods schedule.

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Number	Test/Examination MIL-PRF-32335	Reference paragraph	First Article	Conformance
1	Components missing or not as specified	3.2	X	X
2	Used, rebuilt, or remanufactured components, pieces, or parts used	3.2	X	X
3a	Pressure relief valves for the container is missing or not as specified	3.2.11	X	X
3b	Pressure relief for the RU is missing or not as specified	3.2.12	X	X
4	Seams not as specified	3.2.14	X	X
5	Restraint system missing or not as specified.	3.2.15	X	X
6	Doors missing or not as specified	3.2.18	X	X
7	Anti-pilferage provisions missing or not as specified	3.2.20	X	X
8	Control panel with hour meter and water tight enclosure missing	3.2.23	X	X
9	Power cables/connectors/receptacles missing or not specified	3.2.27	X	X
10	Roof missing or not as specified	3.2.28	X	X
11	Connecting couplers missing or not as specified	3.2.29	X	X
12	Floor missing or not as specified	3.2.30	X	X
13	Refrigeration unit missing	3.2.4	X	X
14	Dimensions not as specified	3.3.1	X	X
15	Weight not as specified	3.4.1	X	X
16	Exterior markings missing or not as specified	3.4.9	X	X
17	NSF certification marking missing or not specified	3.4.8	X	X
18	Approval plates missing or not as specified	3.4.11	X	X
19	Corner fittings missing or not as specified	3.4.3	X	X
20	Forklift pockets missing or not as specified	3.4.4	X	X
21	Tie-downs missing or not as specified	3.4.5	X	X
22	Consolidated/CSC data plate missing or not as specified	3.4.7/3.4.15	X	X
23				
24	Coating thickness not as specified	3.5.11	X	X
25	Color not as specified	3.5.11	X	X
26	Corrosion Control Performance not as specified.	3.4.3	X	X
27	Dissimilar metals are not effectively insulated from each other	4.6.3/6.17	X	X

4.2 Verification Methods. Verification methods can include visual examination, measurement, testing, simulation, modeling, engineering evaluation, component properties analysis, certification, and similarity to previously approved or previously qualified designs. Any verification methods not verified by actual testing shall be pre-approved by the Government.

4.2.1 Verification Alternatives. The contractor may propose alternative test methods, techniques, or equipment, including application of statistical process control, tool control, or cost-effective sampling

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procedures to verify performance (see 6.4). Commercial Reliability and Maintainability (RAM) data may be acceptable at the Government's authority. Any verification alternatives test methods shall be pre-approved by the Government.

4.2.2 Verification Conditions. A uniformly distributed maximum test load is required for verification, unless otherwise specified herein. On completion of each test, the container shall be examined to determine compliance with section 3.

4.2.3 Verification Sequence. The container shall first be tested for weather proofness prior to application of finish painting and/or undercoating. Air tightness and heat leakage testing shall be conducted after all of the ISO structural tests.

4.2.4 Certificates of Compliance. Where certificates of compliance are submitted, the Government reserves the right to check and test such items to determine the validity of the certification.

4.2.5 Payloads. Payloads for mobility tests, unless otherwise specified, shall consist of sandbags or other agreed upon cargo placed on pallets that represents typical cargo configurations used by the Military.

4.3 Operating Requirements Verification. All operating requirements shall be verified as follows:

4.3.1 Air Leakage Test. Conduct an air tightness test on the TRCS in accordance with ISO 1496-2, Test No. 13 – Air leakage test. Verify the maximum air leakage of the TRCS does not exceed the following:

Type I, II or III	7 m^3/hr at 25.4 ± 1mm (1 inch) of water pressure gauge inside (with the refrigeration unit in place)

4.3.2 Heat Leakage Test. Conduct a heat leakage test on the TRCS in accordance with ISO 1496-2, Test No. 14 - Heat leakage test with the RU in place. Verify that the maximum heat leakage of the TRCS does not exceed the following:

Type I, Type II Type III	11.0 Kcal/°C/hr with a mean wall temperature of 293K (20°C) (with the refrigeration unit in place)

4.3.3 Airflow Test and Inspection. Visually inspect the container walls and floor for corrugation or other means to facilitate airflow around cartons of food, and cargo stored in the container. Visually verify that the airflow is not impaired by the cargo and food storage in the container. The interior walls, floor and ceiling shall be designed to permit proper airflow and help prevent the potential of airflow restriction due to cargo loading configuration. Demonstrate that the system operates properly and that proper air circulation is furnished to condition the product. Using thermocouples, verify the TRCS is not recycling rejected heat from the condenser outlet to the condenser inlet. Evidence of rejected heat recycling constitutes a failure. In the same manner, verify that the evaporator is not short cycling and returning freshly supplied air to the evaporator return. Visually inspect that an NSF compliant cage is in place over the evaporator air entrance and exit.

4.3.4 Refrigeration Capacity Tests at 0°F, 35°F at 100°F & 125°F ambient. The refrigeration units shall be sent to an independent testing laboratory to verify and test in accordance with method A4.2 or

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A4.5 described in ARI-1110, (Air Conditioning & Refrigeration Institute) "Standard for Mechanical Transport Refrigeration Units" or other approved Government equivalent. The TRCS shall provide sufficient cooling and heating capacity to maintain internal set point temperatures and cargo temperatures in all ambient conditions listed herein. The refrigeration units shall perform capacity tests at 0°F and 35°F (set point temperatures) inside the TRCS with ambient temperatures of 100°F at 69% Relative Humidity (RH) and 125°F (+0/-3°F) at 3% RH. The capacity tests shall use the methods described in ARI-1110, (Air Conditioning & Refrigeration Institute) "Standard for Mechanical Transport Refrigeration Units". The unit shall be sized properly for above conditions and shall not be too large as to cause short cycling of the refrigeration unit. The contractor shall verify by test and certification that the refrigeration unit possesses a minimum of 50% extra net cooling capacity above the overall heat leakage coefficient of the insulated container at 100°F and maintains set point temperatures with a (+/-3°F) tolerance in all cooling and heating modes. The refrigeration unit shall pull down the interior temperature (both wall and air temperature) of the container from 125°F to 0°F in less than nine (9) hours at 125°F ambient. The refrigeration unit shall also be energy efficient and operate properly in all military transportation and storage operational scenarios.

* The 0°F and 35°F set point thermostat setting temperatures are identified as the return air temperature entering the evaporator and the return ambient air temperatures of 100°F at 69% RH and 125°F (+0/-3°F) at 3% RH entering the condenser.

A certified test report shall be provided to the Government for verification. Data recorded and placed in the test report shall contain, as a minimum, all recorded and graphed data for temperature readings, electrical heat input, suction and discharge pressures, amperage readings, thermal capacity, airflow and complete calculations for the entire design. It shall also contain information describing the airflow parameters and provide clear evidence that no short cycling, or short-circuiting, of the conditioned air will occur. The refrigeration unit shall also demonstrate that it is energy efficient and operates safely without any type of failure, in all military operational scenarios. All panels and doors of the refrigeration unit shall be in the closed position and all accessories and appurtenances shall be in their proper place prior to testing. Recording pressure gages shall be used for these tests. A complete test will be conducted for each power source and temperature listed in Table VA (shown below) in accordance with ARI-1110.

Table VA

Document	Power Supply (+/-10%)	Frequency (+/-3.5%)	Return Air to Air Cooler	Ambient Air
ARI-1110	Multi Fuel Engine Drive		High 35°F Low 0°F	100°F
ARI-1110	208/230 Volt 3-phase	60 Hertz	High 35°F Low 0°F	100°F
ARI-1110	Multi Fuel Engine Drive		High 35°F Low 0°F	125°F (+0/-3°F)
ARI-1110	208/230 Volt 3-phase	60 Hertz	High 35°F Low 0°F	125°F (+0/-3°F)

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The refrigeration unit shall demonstrate that it pulls down to and maintains the interior temperature (walls and air temperature) from 125°F to 0°F in less than a nine (9) hour period at 125°F (+0/-3°F) ambient. Verify the calorimeter calibration is run in accordance with ARI-1110, (Air Conditioning & Refrigeration Institute) "Standard for Mechanical Transport Refrigeration Units. The data shall be recorded automatically by a Data Acquisition Systems and graphed. The data shall include ambient temperatures, internal temperatures, heater temperatures, set point temperatures, suction and discharge pressures and amperage/wattage.

4.3.5 Heating Capacity Test. With the refrigeration unit in the calorimeter, the following procedures shall be conducted.

a. The interior temperature of the calorimeter shall be 30°F ± 5°F.

b. With fans running, the ambient temperature surrounding the condensing section shall be -35° F.

c. A wattmeter shall be attached to read the wattage of the heater.

d. The thermostat shall be set at no less than 45°F.

e. The unit shall be turned on and the wattage reading taken when the calorimeter reaches 40^0F.

The wattmeter shall be properly calibrated and the wattage rating recorded following each test. The wattage measured shall prove that the TRCS refrigeration system is capable of furnishing sufficient supplemental heating capacity to enable an internal temperature of 60°F, with a full load of temperature sensitive cargo, while being subjected to a -35°F ambient temperature. The cargo chosen for this test will be either Unitized Group Ration (Heat and Serve), Meals Ready to Eat (MRE), or a representative product with similar thermal properties that does not require refrigeration. The interior starting temperature shall be 60°F.

4.3.6 Moisture Content and Refrigerant Leakage. Ensure the requirements of 3.2.6 are met.

4.3.7 Defrost Verification and Test. Verify the presence of hot gas or electric defrost system. Verify the RU has provisions for manual defrost (override) initiation and controls that are easily accessible/readable by the operator along with the other common controls. With the refrigeration unit operating in a maximum ambient temperature of 20°F and the insulated structure maintained at 0°F, the evaporator shall be frosted up by spraying (16 ounces) water on the coil surface or evaporating water in the box. After the coil is frosted the refrigeration unit shall be set on the defrost cycle. Verification through testing shall be made that the unit operates through a complete defrost cycle, completely drains all condensate to the exterior and that no overloading of the compressor motor occurs that there is no damage to the compressor because of lack of lubrication or refrigerant "slugging," and that no condensate gets into the interior of the insulated container.

Verify the RU has a defrost cycle terminated by temperature and automatically initiated on demand (when ice builds up on the evaporator coil) or shall automatically defrost the evaporator coil by measuring the difference between the return air temperature and the saturated inlet air temperature. Alternatively, the defrost can be initiated by a time control adjustable from 2 to 12 hour intervals and a

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user selected timed defrost termination is acceptable. The defrost system shall have the capacity to remove all condensate build up on the evaporator coil within 30 minutes. Refrigeration units shall have a drainage system for the collection and discharge of condensate to the outside of the insulated container without pooling of the fluids in the lines. Condensate drain lines shall prevent the infiltration of insects into the lines. The drain line assembly shall utilize commercially available components that are easily disconnected and reinstalled during maintenance and removal of the RU from the IC. The condensate drain assembly shall not damage any rations, equipment, or stored items located in the container. The condensate drain hoses and associated fittings, whether internally or externally mounted, shall be physically protected from damage and from freezing temperatures. The RU shall prevent freezing of condensate in the drain pan and condensate drain hose assemblies.

4.3.8 Safety Certification. Verify the TRCS meets the requirements listed in section 3.2.8. The container shall be certified that it complies with Occupational Safety and Health Administration Standards (OSHA) 29CFR 1910 safety requirements, and that the insulation when subjected to flame shall extinguish within 60 seconds after removal of the flame. Verify that the TRCS is safe for the intended purpose and minimizes flammability, operational and personnel hazards. All potential, operational and personnel hazards shall be identified to the end user. All controls shall be permanently labeled and traceable to the Technical Manual. All Electrical and Electronics used in the TRCS shall be permanently marked for identification for proper and safe field usage.

4.3.8.1 Container Certification. A Convention for Safe Containers (CSC) and TIR plates shall be applied for and obtained from a designated approval authority or classification society. They shall be attached and displayed on the container as required by the convention in accordance with Department of Transportation (DOT) CFR 49, parts 450 and 451 and provided to the Government. The contractor shall certify that the TRCS meets the dimensional requirements of ISO Standard 668 and ISO Standard 1161 and the structural and thermal requirements of ISO Standard 1496-1 and 1496-2. The contractor shall also provide a written Certificate for Safe Containers from the Coast Guard for the TRCS.

4.3.8.2 Wiring Certification. Wiring practices used in the refrigeration units shall be certified as conforming to NFPA Standard No. 70 by submission of manufacturer's certification, certified test report, or appropriate labels or listing marks.

4.3.9 Uncontrolled Hazards Inspection. Verify that the TRCS presents no uncontrolled hazards to operators or poses any potential damage to Government equipment. Verify that a fail-safe design has been utilized for critical safety/health function components or subsystems. Verify safeguards designed into the system, and ensure that a clear indication is provided when any unauthorized tampering has taken place within the system. Ensure the requirements of Section 3.2.9 are met.

4.3.10 Floor Drainage Test and Inspection. Visually inspect the container for drains. Verify the water tightness with drains in the closed position by completely filling the floor with water to a depth not less than 1 inch. Duration of the test shall be not be less than 2 hours. Upon completion of test, the floor should be free of standing water. Verify automatic release by inducing an internal pressure above normal operation pressure inside the container until the drains automatically release. All loose hardware shall be restrained using lanyards. Demonstrate the operation of the drains are not negatively affected by sand that could be tracked or blown into the container in sandy/dusty environments.

4.3.11 Pressure Relief inspection and test. Verify the presence of an "automatic pressure relief mechanism" and an emergency/air pressure relief hatch in the right hand door. Test the TRCS to meet

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the rapid decompression requirement of MIL-STD-810, Method 500.0, procedure I. Verify the hatch is operable from the interior of the container and allows no air/water intrusion when in the closed position during all testing listed herein.

4.3.12 High Pressure Cutouts Demonstration. Ensure the requirements of 3.2.12 are met. Demonstrate the presence of high pressure cutout switch. Block the condenser and demonstrate the unit shuts "off" at high head pressure. Measure and document the cut out high and low pressures on the technical manual.

4.3.13 Heater High Limit Cutout. Ensure the requirements of 3.2.13 are met.

4.3.14 Seams Inspection. Ensure the requirements of 3.2.14 are met.

4.3.15 Cargo Restraint System Verification. Verify the presence of at least 12 tie-down points inside the TRCS container. Verify the tie-down points are fully compatible with special purpose tie downs straps (NSN 3990-01-304-3009). Verify for Type I, there are four tie downs on each of the lower portions of the front and rear (8') walls of the container and two tie downs on the RU and door side walls/floors. Verify the tie-downs are positioned so that there are four tie downs on each side of the lower portion of the container. Verify for Type III, the two tie downs located on the door side shall be mounted on the lower portion of the walls. Verify any tie downs located on the floor are fully recessed into the floor. Demonstrate the tie-downs are capable of accepting one-inch wide metal strapping without tearing the strapping. Verify adequate clearance is provided at the tie-down for installation of the metal hook of adjustable web strapping. Verify by inspection of the drawings that non of the tie-downs create a direct thermal short to the outside of the container and can withstand a minimum 1,000 lbs load at a 45-degree pull angle without permanent deformation. Verify tie-down provisions shall restrain the load without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. The contractor shall provide data as evidence to meeting the tension requirements listed above. Verify each TRCS is supplied with a minimum of seven (7) "special purpose straps ((web, tie down, NSN: 3990-01-304-3009, PN: FDC5770-5 (Cage Code: 98313)). The interior of the TRCS container shall also be compatible with the mounting or and the rated loads for the shoring beam (NSN: 9540-01-491-3804). Verify the interior of the container is supplied with a minimum of two (2) A and E-tracks per the illustration listed to meet the critical equipment pack out requirements.

4.3.16 Vinyl Strip Inspection. Ensure the requirements of 3.2.16 are met.

4.3.17 Technical Manual Holder Inspection. Verify the presence of a waterproof manual holder that is fastened to the container and that can be opened or closed without the use of tools. Verify interior dimensions as being not less than 12 inches by 9 inches by 1.5 or 3 inches with deep Black lettering reading, "TECHNICAL MANUAL HOLDER" permanently stenciled on the exterior of the manual holder.

4.3.18 Door Configuration Inspection. Visually inspect that the container has two doors that are hung within the end frame and are in accordance with ISO 668. Verify that the doors are secured in such a manner so as to prevent access to the interior of the container without leaving visible signs of tampering. For Type I and II, verify that each door is provided with anti-rack provisions. Verify that all moving parts of the door locking mechanism are permanently lubricated and that there is a means to hold doors in the full open position so that they do not scrape or chafe the container. Verify that the door gasket material can be satisfactorily used at -40 °F to 125°F. Verify that the door gasket material

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maintains a tight seal and performs properly within all conditional ranges of both hot and basic climatic conditions without any adverse effect on the performance, components, and materials. Verify that gaskets and door seal does not impede door opening and closure. The exterior of the TRCS and the doorjambs including gasketing shall show no visible evidence of ice build up and condensate at the set point temperature of 32°F or less at an ambient of 80°F or higher. While maintaining set point temperatures, any sign of ice build up and visible condensation on the doorjamb, the gasketing, or on the exterior of the TRCS constitutes a failure.

4.3.19 Door Security Inspection. Inspect the door closing devices on the container doors to verify when in the closed position and when padlocked they prevent access to the interior of the container. Verify that the door can properly open and close without binding. Verify the presence of pull handles to facilitate door opening. Verify that all locking device handles are furnished with provisions for padlocking and customs sealing according to ASTM F883. Verify that no door, hardware, or locking components extend beyond the ISO corner fittings or overall ISO dimensions. Verify that the escape means has luminescent strips to direct an individual to the egress. Inspect the door closing devices on the container doors that when in the closed position and padlocked, prevent access to the interior of the container. Verify that the door can properly open and close.

4.3.20 Anti-Pilferage Inspection. Ensure the requirements of 3.2.20 are met.

4.3.21 4-Hour Operation Test. The refrigeration unit evaporator shall be mounted on the TRCS container. All refrigeration unit components shall be in place and the container doors shall be closed. The refrigeration unit shall be operated from both the electric source and/or the multi fuel engine source for 4 hours to a steady state condition to determine that all mechanical and electrical components are functioning as intended and all controls are in calibration. The units shall have a heat load applied to the enclosure so that the unit operates more than 50 percent of the test time and cycles on the automatic controls three or more times per hour. Suction pressure at the compressor shall be verified to be within the established guidelines of the compressor manufacturer. Prior to completion of the 4-hour test, the air flow to the condenser shall be blocked off and the unit operated until automatic shutdown of the refrigeration unit occurs to verify the safety device shuts the compressor off when the manufacturers compressor discharge limit is exceeded. During normal operation, the discharge pressure shall be observed to be not less than 100 pounds per square inch gage (psig). The TRCS shall also properly defrost, and remove condensate.

After a minimum operating time of 1 hour with the thermostat set at 0 °F the moisture indicator shall be observed to verify that the moisture content as shown by the component manufacturer's color chart data as is reading dry. A check for refrigerant leakage shall also be performed with a leak detector having a maximum sensitivity of 0.40 oz per year. Fory Type II, leakage shall be checked to verify that it is no greater than 0.5 oz per year, except the compressor shaft seal leakage is no greater than 1.0 oz per year. No visible condensate is allowable from the exterior and in the doorjamb during this test. At the start and after 3 hours, the nominal amperage shall also be recorded to verify it is within design limits. Temperatures, pressures and amperage/voltage shall be automatically recorded and graphed.

4.3.22 Temperature Indicator Inspection. Visually inspect and test the temperature control to verify that temperatures from minus 15°F to plus 80 °F can be selected in increments of 1°F with a set point tolerance of +/- 3°F. Verify that the temperature indicator is readable in all modes of operation and in all of ambient conditions listed herein. Verify the TRCS temperature indicator is reading the proper temperature by comparing data from an external (calibrated) temperature measuring device.

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4.3.23 Temperature and Control Panel Test and Inspection. If a remote control panel is provided, visually inspect the control to verify that is it mounted in a NEMA enclosure. Demonstrate during the 4 hour operational test and the design reliability testing that the TRCS controller properly performs the following functions in ambient temperatures ranging from –25°F to 125°F (+0/-3°F).

n) Turn the refrigeration unit "ON" and "OFF.
o) Change and adjust the interior set point temperature. (i.e. thermistor to monitor/control internal container temperature). Identify the interior temperature.
p) Initiate a manual defrost function.
q) Indicate when the power is "ON" and "OFF" (LED indicator or lit LCD display indicating the presence of AC power)
r) Identify alarm conditions. (If an alarm light or alarm code is present, the system shall identify if a problem exists and allow the user to take corrective/troubleshooting actions.)
s) Frozen range lockout, which prevents the user from inadvertently damaging the load by lowering the set point temperature.
t) Main power supply with manual re-settable circuit breaker (properly sized and rated) to protect the refrigeration system.
u) Hour meter (waterproof, weather tight analog or digital which reads compressor run time)
v) Set point range indicator -15ºF to +80ºF with minimum increment of 1ºF with a control tolerance of ± 3ºF. For Type II an engine hour meter is also required.

For Type II, verify the presence and proper operation of the following controls and indicators:

a. Battery Charge Level Indicator
b. Fuel Level Indicator
c. Fuel level warning (for low fuel level)
d. Automatic Reset Switch/ Engine Protection
e. Battery fault indicator
f. Fuse diagnostic indicator
g. High pressure refrigeration system fault indicator
h. Water/oil safety fault indicator
i. High/low temperature fault indicators for refrigerant circuits (both audio and visual with on/off switch)
j. Manual Preheat of engine driven refrigeration unit operational indicator
k. Operation of refrigeration unit operational indicator
l. Power saver On/Off of engine driven refrigeration unit operational indicator
m. Battery saver (prevents battery drain beyond the point that prevents restarting without re-charging the battery)

Demonstrate that the selected temperatures are controlled automatically and allow an incremental selection of normal refrigerator/freezer temperatures for all ambient climatic conditions. Verify lockout of heating cycle if required. Demonstrate that the TRCS temperature control system capable of controlling the temperature in the TRCS at 35°F +/-3°F and 0°F +/-3°F.

4.3.24 Ozone Depleting Chemicals Inspection and Certification. Visually check the list of prohibited Class I and Class II ozone-depleting substances on the Environmental Protection Agency (EPA) website. The systems shall not contain or be manufactured with any of these substances. The

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contractor shall provide certification to meeting the EPA requirement listed in Section 3. The contractor shall certify the refrigerant charge quantity on the refrigeration data plate.

4.3.25 Refrigeration Service Valves. Verify the presence of high and low side service valves, service port gauge connections and valve caps. Verify the valves can be "front seated" or "back seated" as necessary using the appropriately sized refrigeration service valve wrench. Verify each valve does not leak at the connection to the compressor, the valve steam, the port connections or any other location. Verify the port connections and the main valve stems have metal vapor proof caps that are accessible using the appropriately sized open end wrench.

4.3.26 Power Test and Inspection. During the 0°F and 35°F set point temperature portion, at an ambient of 125°F (+0/-3°F), observe that the refrigeration unit operates as intended in the following modes:

1. For Type II, via multi fuel engine driven refrigeration unit to steady state condition according ARI-1110 plus 2 hours of net cooling operation with one defrost cycle completed.
2. On a 208 Volt (+/-10%) 3-phase 60-Hz (+/-3.5%) power source operate the unit to steady state condition according ARI-1110 plus 2 hours of net cooling operation with one defrost cycle completed.

The TRCS shall be tested at the above temperatures, voltages and frequencies. Verify that the TRCS starts and operates properly, and safely, without any physical damage or degradation to system performance. Voltage, current, frequency, temperature and time data shall be recorded and any anomalies shall be identified. The TRCS unit shall be capable of being operated properly and safely while in transit or stationary. All parts of the electrical system shall comply with UL, NEMA, ISO and NFPA Standard No. 70.

4.3.27 Power Supply Cables and Connectors Certification, Inspection and Demonstration. Option 1 (Army): Verify the refrigeration unit is supplied with a power cable. Verify one end of each power cable shall be permanently attached (hardwired) to a weather proof, fixed mount power input connection box, or directly to the refrigeration system unit. Verify the other end of each power cable is supplied with the proper 60Amp, Class L (MIL-DTL-22992), Connector Plug, Straight, attached to the free end of a suitable 12 foot SOO type cable. The Power input point shall have a weatherproof circuit breaker installed between the power input connection and the refrigeration unit Ensure the connector has the suitable mating plug or cap that is attached to the end of the cable to protect it from the environment when not connected. All power connections and cables shall be heavy duty, water tight, corrosion resistant, strain relieved and protected from physical damage. The contractor shall certify the power plugs are easy to install, ready to connect, maintenance free and rated for a 20-year service life.

The power cables, connectors and attachment points shall be constructed of non-flammable material, shall be compliant with UL, NEMA, ISO and NFPA Standard No. 70, shall possess proper current rating and grounding and be robustly designed to withstand the rigors of transport and operation (this is to include, but not limited to, breaking/tensile strength, abrasion resistance, oil resistance, waterproof, weatherproof, dustproof and corrosion resistance). Verify the power cable is from the type SOO family.

Verify all power cables/connectors shall be heavy duty, strain relieved and restrained from movement, where necessary. The power connectors shall utilize impact resistant housings and provide impact

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resistance in temperatures ranging from –25 °F to 130 °F. The power connectors shall utilize heat resistant contact carriers. The entire power cabling and electrical system shall be rated to handle the current draw of the TRCS in all operational scenarios

Electrical enclosures (junction boxes, control boxes, etc.) shall be certified as conforming to the water tightness requirements of NEMA MG-1 or 4X by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks. Verify the refrigeration unit has the capabilities to operate on external power supplied by the standard DOD family of generators and commercial power sources. Verify all cabling and wires are tagged and/or color coded and traceable to an electrical schematic. Verify the entire power cabling and electrical system is rated to handle the nominal and peak current draw of the refrigeration unit in all operational scenarios. Verify each high voltage power connector shall be stenciled or labeled with the appropriate voltage and amperage rating. Verify the TRCS has provisions for safely storing the power supply cables/connectors within the TRCS ISO dimensions while stationary and in transit.

Option 2 (Air Force): As specified on the contract, verify the TRCS shall is supplied with a box mount, 45-degree, 60 Amp, Class L receptacle per MS90558-32412P (Mil-DTL-22992) hard wired to a weather proof, fixed mount power input connection box, or directly to the refrigeration system. Verify the receptacle is provided with the mating plug and lanyard chain.

Note: Verify that options 1 and 2 shall allow the RU to operate properly and safely using 208/230 volt, 3-phase, 50/60 Hz power and shall be suitable to handle the refrigeration unit’s amp draw. Verify the incoming voltage and current is stenciled on the connector or it’s housing in legible black or white lettering. Verify the presence of a high voltage placard/label placed on the incoming power connector. Verify all the TRCS RU wiring configuration is outlined in the documented wiring diagram and schematic. Verify all power connections and cables shall be heavy duty, water tight, corrosion resistant, strain relieved and protected from physical damage. Verify the entire power cabling and electrical system shall be rated to handle the current draw of the TRCS in all operational scenarios. Verify all power supply cabling shall be safely stowed and contained within the ISO clearance allowance dimensions of the system. For Type II only, demonstrate the TRCS prevents the onboard engine driven refrigeration unit and an external power source from simultaneously powering the electrical system.

4.3.28 Roof Test and Inspection. Demonstrate that the TRCS is capable of withstanding a uniform load equivalent to 40 pounds per square foot across the entire roof surface without any detrimental effect or damage to the container or any other TRCS component. The contractor shall certify and demonstrate that the roof of the TRCS can safely support the weight of personnel and equipment necessary for all set up, tear down, operation, slinging and maintenance tasks which take place on the roof without evidence of damage or degradation to any of the containers constituents. This can be accomplished during the static lift testing or during the external air helicopter lift. Visually inspect the roof to verify that it is self-draining, that it prevents water pooling, and prevents water from both entering the container and/or causing component damage following exposure to actual and/or simulated rainfall. Verify that a non-slip surface has been applied. The contractor shall provide certification and demonstrate that the TRCS has roof access steps/handholds that are permanently attached to the exterior of the container for safe access to and from the roof. Verify that each step/handhold is located within the ISO dimensions. Certification shall be furnished that each step/handhold is capable of supporting at least 400 pounds per step/handhold without damage to the TRCS.

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4.3.29 Connecting Couplers, Quantity and Storage Inspection. Three (3) TRCS containers, dry freight TriCons or any combination shall be coupled together to form a series 1C equivalent 20' module container. The currently used coupling NSN is 3040-01-387-4048. Three coupling shall be supplied per TRCS container. The couplers shall be supplied in a fiber board box, labeled and placed inside the production TRCS units.

4.3.30 Floor Test and Inspection. Verify that the floor is watertight and is permanently attached to its bottom side rails and cross members. Verify that the floor can support the container's payload (P_g) as defined in ISO 1496-1, paragraph 6.1.1.

4.4 **Interface and Interoperability Requirements Verification**. All interface and inter-operability requirements shall be verified as follows:

4.4.1 Dimensions Inspection. Measure and verify that the dimensions, tolerances and diagonal differences of the container meet the requirements specified in ISO 668 for type 1C containers.

4.4.2 Refrigerator Interface Inspection and Certification. Demonstrate that the refrigeration unit interfaces properly with the container and a vapor/air tight seal is made between the container and refrigeration unit. The end wall, opposite from the end wall with doors, shall demonstrate that it possesses an integral refrigeration unit capable of maintaining temperatures as low as 0ºF in environments as described in this specification. An integral electric driven refrigeration unit shall be incorporated into the container design and verified by inspection and testing. Verify that the amount that the refrigeration system extends into the container is minimized to allow the internal useable volume to be maximized. Verify that the interface has an opening for mounting a refrigeration unit. The end wall opening shall be verified to exhibit sufficient strength to support the refrigeration unit. The contractor shall certify that the end wall opening is of sufficient strength so as to support a refrigeration unit weighing 500 pounds. Demonstrate that the mounting capability of the TRCS is backwards compatible with the Thermo King VM405 (Model PBBDD2) refrigeration unit.

4.4.3 Padlock inspection. Verify that the padlock and keys are as specified in ASTM F883 and fits in the latch on the door handle. At least two keys shall be furnished for each padlock.

4.4.4 Interchangeability and Ease of Maintenance Certification. The contractor shall certify and verify that all component parts of the container, which are replaceable, are also interchangeable with the same parts of other containers furnished under the same contract, and that all assemblies and installed attachments are accessible for preventive maintenance procedures. Verify that all assemblies and installed attachments are designed for ease of installation and replacement. Verify that they are accessible for preventive maintenance actions and that they do not require the removal of other major assemblies and installed attachments.

4.4.5 Interior and exterior light test and inspection. For Type III verify the presence of a vapor and shock proof lighting fixture or fixtures mounted inside the interior of the IC. Verify the presence of a protective cover, which is shatterproof and removable without special tools for bulb replacement. Verify all wiring is shielded from physical damage in electrical conduit and conforms to NEC standards. Verify all conduit and lighting components prevent contact with other interior components and can be easily accessible for maintenance, repair or servicing. Verify all conduit and lighting components are designed for long life and can withstand extreme cold, heat and thermal shock without degradation. Verify by testing the lighting system can withstand all of the operational, environmental

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and transportability requirements listed herein without failure. Verify the fixture shall accept and be supplied with a minimum 100 watt, 120 VAC incandescent light bulb(s) or LED lighting system with equivalent lumens. Verify the presence and operation of a switch for the light located inside and adjacent to the door and shall be vapor tight and protected from damage. Verify the fixture is installed in a location to prevent light obstruction. Verify and test the interior lighting system operates with full blackout mode capabilities. Verify the interior light shall be designed to plug into the weather tight exterior grade 15 Amp, GFCI duplex outlet mounted on the exterior of the container.

Verify the presence of a heavy duty portable exterior light system (PELS) with mounting bracket is supplied with and over packed with each TRCS (Type III). Verify the PELS attached to and can be removed from the exterior of the container without special tools and shall provide focused light on the door side of the container to support safe access/egress/loading and unloading by personnel in no or low light conditions. Verify the PELS shall be supplied with a minimum 25 foot, 15 Amp, 3 prong 120VAC power cord that interfaces with the weather tight GFCI duplex outlet on the exterior of the container and can be powered from alternate 120 VAC, 60 Hertz shore power. The mounting bracket for the PELS shall be supplied with an integrated weatherproof/watertight light switch that is located near the door entrance and less than 6 feet high from the ground. The PELS shall be designed to withstand the rigors of repeated field use without degradation or damage. Each PELS shall be supplied with one spare bulb. When deployed, the PELS shall be subjected to and pass all of the operational tests listed herein. The PELS shall be by subjected and pass all of the operational tests listed herein. When stowed, the PELS shall be subjected to and pass all of the structural and transportability tests listed herein.

4.4.6 Human Factors. Verify and demonstrate that the dimensions and layout of the TRCS permit safe and effective operation by male and female personnel in the 5th to 95th percentiles while wearing the field duty uniform, mission-oriented protective posture (MOPP) level-II, and the cold weather ensemble during all potential environmental and operational conditions. Relevant anthropometrics measurements are listed in Table II, and additional guidance can be found in the 1988 Anthropometrics Survey of U.S. Army Personnel (see 6.10).

TABLE II. Anthropometrics dimensions.

Description	Dimension to be used	Example application
Overhead reach	73 inches (5th percentile female)	Shelf height
Functional reach	32 inches (5th percentile female)	Counter depth
Stature	74 inches (95th percentile male, boots)	Head clearance
Shoulder breadth	27 inches (95th percentile male, bulky clothing)	Passage clearance
Weight	199 pounds (in BDU)	Roof weight limits

4.4.7. Test of the performance of a container with refrigeration by a mechanical refrigeration unit (MRU) Test. The TRCS shall undergo and pass the performance of a container with refrigeration by a mechanical refrigeration unit (MRU) Test per ISO 1496-2, Test No. 15a.

4.5 Support or Ownership Requirements Verification. All support or ownership and requirements shall be verified as follows:

4.5.1 Weight and CG Inspection. Verify the TRCS weighs less than the following:

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Maximum Gross Weight (R1)	11,090 kg	(24,450lbs)
Tare Weight (Max. including the refrigeration unit) (T1)	2,000 kg	(4,410 lbs)
Payload (Min.) (P1)	9,089 kg	(20,040 lbs)

For Type I and III the weight of the refrigeration unit shall not exceed 500 lbs. Verify the presence of securely mounted lifting provisions and/or forklift pockets on the refrigeration unit.

4.5.2 Structural Requirements Tests. The structural requirements shall be verified as follows. Although the TRCS is not considered a 1DX (10') container module, the tests on a single TRCS shall meet 1DX structural requirements.

4.5.2.1 Stacking Test. The containers shall undergo the stacking tests of ISO 1496-2 Test No. 1 – Stacking as an ISO 1DX (3 high) and 1C (9 high) containers. Upon completion of tests verify that the TRCS operates correctly and exhibits no permanent deformation or abnormality.

4.5.2.2 Lifting from the Four Top Corner Fittings Test. The containers shall undergo the lifting test of ISO 1496-2 Test No. 2 - Lifting from the four top corner fittings as a 1C (can use other military approved TriCons) and a 1DX (to a single point at a slinging angle of 60 degrees) container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality. This test shall be performed on one container and three containers coupled together. When conducting the 1C lift test, the lifting forces shall to be applied vertically from the top corner fittings and the test payload = (2(R)-T). When conducting the 1DX lift test, the lifting forces shall be applied vertically from the horizontal and the test payload = R=22,400 lbs.

4.5.2.3 Lifting from the Four Bottom Corner Fittings Test. The containers shall undergo the lifting test of ISO 1496-3 Test No. 3 - Lifting from the four bottom corner fittings as a 1C (can use other military approved TriCons) and a 1DX container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality. This test shall be performed on one container and three containers coupled together. When conducting the 1C lift test, the lifting forces shall to be applied at a 45 degree angle from the horizontal bottom corner fitting and the test payload = (2(R)-T) (When R = 52,900 pounds). When conducting the 1DX lift test, the lifting forces shall be applied at a 60 degree angle from the horizontal and the test payload = (2(R)-T) When R=22,400 pounds.

4.5.2.4 External Restraint (longitudinal) Test. The containers shall undergo the restraint test of ISO 1496-2 Test No. 4 - External Restraint (longitudinal) as an ISO 1C container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality.

4.5.2.5 Strength of End Walls Test. The containers shall undergo the test of ISO 1496-2 Test No. 5 - Strength of end walls as a single ISO 1DX container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality. Test payload (ISO): Payload (P1) * (0.6*P).

4.5.2.6 Strength of Side Walls Test. The containers shall undergo the test of ISO 1496-2 Test No. 6 - Strength of sidewalls as an ISO 1DX container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality. Test payload (ISO): Payload (P1) * (0.6*P).

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4.5.2.7 Strength of the Roof Test. The containers shall undergo the test of ISO 1496-2 Test No. 7 - Strength of the roof as an ISO 1DX container. Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality. Test payload 660 lbs. over a 24" x 12" area (i.e. 330 lbs/ft^2)

4.5.2.8 Floor Strength Test. RESERVED

4.5.2.9 Lifting from Forklift Pockets Test and Inspection. The container shall undergo the test of ISO 1496-2 Test No. 11 - Verify the container base contains a set of four (4) way forklift pockets/tubes that are a minimum of 4.0" high, 12.0 "inches wide, and conform to ISO 1496-1, Annex C. The minimum test payload shall be 1.6R. The container shall be able to meet this requirement when lifted and transported by the forklift with 42 inch or 57 inch long tines. The base of the container shall be supplied with reinforcement to withstand this requirement without damage or permanent deformation.

4.5.2.10 Container Coupling Test. Three (3) TRCS containers and/or TRCS/dry freight TriCon container combinations shall be coupled together to form a series 1C equivalent 20' module container. The connecting couplers shall provide a connection between the vertical end faces of the corner fittings of each container. Four identical connection couplers shall be used at each container interface, two at the top and two at the bottom. All connecting couplers shall be fully assembled and all parts shall be capable of being operated without special tools. Each connecting coupler shall be capable of being securely fastened to a single container's corner fitting allowing the connection to be made first to one container and then to another. In a coupled 1C configuration, and when the containers are loaded to their maximum gross weight, they shall be tested and show no visible signs of permanent deformation after being subjected to the top lift, bottom lift, and forklift testing listed above. The currently used coupling NSN is 3040-01-387-4048. Three coupling shall be supplied per TRCS container.

4.5.2.11 Rigidity (transverse) Test. The container shall undergo the test of ISO 1496-2 Test No. 9 - Rigidity (transverse) as a 1C container (both sides). Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality.

4.5.2.12 Rigidity (longitudinal) Test. The container shall undergo the test of ISO 1496-2 Test No. 10 - Rigidity (longitudinal) as a 1C container (both sides). Upon completion of test verify that the container operates correctly and exhibits no permanent deformation or abnormality.

4.5.3 Corner Fittings Inspection. Verify the corner fittings conform to the dimensional and strength requirements of ISO 1161. Verify the corner fittings meet durability requirements during all loading, interfacing and environmental testing specified herein. All externally mounted hardware including the RU shall be within the inside plane of the ISO clearance allowances.

4.5.4 Forklift Pocket Test and Inspection. The contractor shall demonstrate that the TRCS has four (4) way forklift pockets to enable the container to be safely lifted, moved, and placed as needed by both military and commercial forklifts. The TRCS shall be tested to assure it can be safely carried in all circumstances in the unloaded (TARE weight) configuration. A visual inspection and an operational checkout shall be conducted prior to and after the forklift tests. The TRCS shall be inspected internally and externally after each movement for damage. Any physical damage to the TRCS, or the forklift during or after the forklift tests shall constitute failure of these tests. Should any dimensions and tolerances result in a 60-inch component or assembly not accommodating the tine ways, then the TRCS

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shall be judged as not meeting the requirements of this MIL-PRF and failure of the FAT. Reinforcements and/or puncture plates shall be provided around the openings to prevent damage to the TRCS when the forklift tines are engaging or disengaging the TRCS. Corrosion resistant material shall be applied to the inside of the forklift tubes. Verify label presence and wording above pockets.

Fork Lift Pocket requirements: The fork lift pockets shall be visually confirmed with dimensional measurement for conformance with Army and Air Force material handling equipment required dimensions.

4.5.5 Multi Fuel Engine Driven Refrigeration System Demonstration (Type II only).
Verify/demonstrate the refrigeration system has an integrated multi-fueled (JP-8 and diesel) engine driven system. Verify the engine driven section is an integral part of the refrigeration unit (See 6.2) and demonstrate that it is provided with a drain valve to fully drain the fuel tank in the integral engine-driven system. Demonstrate the TRCS has the capacity to generate sufficient power to start and continuously supply the refrigeration system, all container controls, additional fans/ evaporators, and any other powered devices that are part of the TRCS. The TRCS shall be provided with switching device capable of changing from integrated engine driven to an external electrical power source.

The engine selected shall have the following features:
a. Coolant level gauge
b. 50/50 anti-freeze
c. Radiator coolant reserve
d. Heavy-duty coolant hoses
e. Engine coolant high temperature cutout
f. Engine low oil pressure cut-out
g. Power switch with multiple positions for each of the power sources referenced and an OFF
h. Main Power Switch
i. Battery
j. Auxiliary Fuel Tank

The engine driven RU control panel shall have the following features:
A. Battery Charge Level Indicator
B. Fuel level indicator
C. Fuel level warning (for low fuel level)

Verify the TRCS is provided with an auxiliary fuel tank to store and provide fuel for the refrigeration unit when operating in engine driven mode. The contractor shall provide a DOT certified fuel tank capable of storing enough fuel to keep the refrigeration operating at 100% capacity, during the worst ambient conditions specified herein for at least 24 hours without refueling. Verify the tanks interior surface are permanently protected from corrosion. Verify the tank includes a protected fuel gage that is clearly visible to the user. Verify the tank includes; baffle plates as required, essential pressure relief valves, filler cap, drain valve and a captive retained inlet fuel screen. Demonstrate the drain valve allows the tank to be completely drained prior to transport. Demonstrate these devices prevent fuel spillage when the container is operated up to a maximum of a 20 percent slope/grade in all axis. Verify that a highly durable, abrasion resistant fuel line is provided which transfers fuel between the auxiliary fuel tank and the engine driven RU. Verify a captive retained cap for the fuel line fitting is provided. Demonstrate the auxiliary fuel tank is attached to the container frame in a manner to facilitate easy removal and shall be properly grounded. Verify that provisions shall be made for storage of any excess

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length of multi-fuel engine driven RU auxiliary fuel line near the auxiliary fuel tank. Verify the auxiliary fuel line shall be installed/located in such a way so as to minimize and eliminate operational and safety hazards. Verify that electrical and fuel lines do not be run adjacent to one another. Demonstrate there are no hazards to include, but are not limited to, electrical, mechanical vibrations, abrasion, puncturing, and environmental issues such as exposure to UV radiation. Verify the type and quantity of fuel is permanently stenciled onto the fuel tank

4.5.6 Roof Load and access/steps/handholds inspection/demonstration. Demonstrate the roof is self-draining, prevent water pooling, and prevents water from both entering the container and/or causing component damage. Demonstrate the roof (per ISO 1496-1) does not deflect more than ¼ inch permanent deflection. When the TRCS is tested in accordance with 1496-1, Weatherproofness test, and is level to within 5 degrees, the roof shall retain not more than 1/8 inch of water. Demonstrate the roof of the Insulated Container shall support the weight of all personnel and equipment necessary to complete set up, tear down, operational procedures, rigging procedures and maintenance tasks which take place on the roof without evidence of damage or degradation. Demonstrate the roof shall be capable of supporting, without damage or degradation; (1) a uniformly distributed load of 40 lbs/ft^2 over the entire surface and (2) a concentrated load of 660 lbs over a 1 ft x 2 ft area, applied anywhere on the roof.

Demonstrate the TRCS shall have safe roof access steps/handholds enabling ground level personnel to safely access the roof level of the container. Demonstrate each step/handhold shall be capable of supporting at least 400 pounds and shall not extend beyond the container footprint. All steps/handholds shall be permanently attached to the exterior of the container and shall comply with transportation requirements.

4.5.7 Data Plates and Marking Inspection. Verify the presence of a data corrosion resistant metal identification plate permanently affixed to the container, with the required data (to include interior and exterior marking) as required.

(Note: Tare weight. The tare weight marked on the container shall be the weight of the container including the weight of the refrigeration unit.

TriCon Refrigerated Container System (TRCS)
Performance Specification:
NSN: (see 6.2)

Serial Number:
Tare Weight, Equipped: LB_____ KG_____

Contract Number:
MFD By:
Date (Month & Year)
Technical Manual No.

4.5.8 Sanitary Inspection. Visually inspect the interior of the container to verify that there are no open seams, pits, narrow crevices or any areas, which cannot be sanitized by conventional cleaning methods using steam, hot water, pressure washers and detergents. Verify that no materials inside the container are adversely affected by such sanitizing methods as steam cleaning, pressurized power

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washers, bleaching agents, mild detergents or detergents normally used. The TRCS interior shall be stainless steel and designed and manufactured in accordance with NSF Standard No. 2, C-3 and 7. Cleaning procedures and cleaning agent materials shall be documented in the technical manual. The interior of the TRCS shall be NSF Standard No. 7 certified and an up to date written listing shall be provided to the Government and kept on file for future Government inspections.

4.5.9 Exterior Marking Inspection. Verify that all markings are permanent and certified to remain clearly visible/legible throughout the entire life of the TRCS. Verify that the container is identified and marked with black paint in accordance with ISO 6346 and TB-43-0214 with the following additional requirements:

a. "CENTER OF BALANCE" shall be stenciled at the longitudinal center of balance of the empty TRCS. The container shall be stenciled on both sides with a 4-inch high vertical arrow and lettering 2 inches high at the longitudinal center of balance.

b. "JP-8/DIESEL ONLY" shall be stencilied on the auxillary fuel tank in lettering 2-inches high.

c. "TECHNICAL MANUAL HOLDER" shall be stenciled on the manual holder in black lettering.

d. The following lettering with applied appropriate weights and data shall be stenciled on the exterior door of the TRCS.

MAX.GROSS WEIGHT	Kg and lbs
TARE WEIGHT	Kg and lbs
MAX PAYLOAD	Kg and lbs
CUBIC CAPACITY	M3/CU.FT.
NSN:	
LIN:	

e. The containers "vertical height" shall be stenciled in both meters and feet with a 4-inch minimum height lettering on the exterior of the container.

f. Any other safety markings, required for warning personnel of potential safety hazards, that may be encountered upon removal/relocation of partitions, barriers, guards and/or doors.

4.5.10 Interior Marking Inspection. Verify that the container serial number is stamped, or stenciled, on the interior surface of the top rail or corner fitting at the door end. Verify that the following is stenciled in 1-inch high black lettering on the interior surface under the evaporation exit:

"WARNING: DO NOT BLOCK AIR CIRCULATION."

Verify the presence of permanent markings located inside the TRCS container to identify where cargo cannot be stored, so as to facilitate proper airflow.

4.5.11 RESERVED

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4.5.12 Instructions and Warning Placards Inspection. The TRCS will be visually inspected to assure that all of the appropriate warning labels and instructions to assist personnel with safe operation and emergency procedures are in place. Placards will be inspected to assure that they are permanently affixed to equipment, and will be sufficiently durable to remain legible at all times throughout the entire systems life. Placards will be visually inspected to determine if they are easily noticeable, recognizable, and understandable. Permanent metal wiring diagrams and electrical schematics shall be mounted to the TRCS in a location accessible and readable to the service technician

4.5.13 Refrigeration Unit Identification Data Plate. Verify that all placards are noticeable, recognizable, and understandable. It shall contain the following information:

1. Contract Number
2. NSN Number
3. Manufacturers Name
4. Manufacturers Address
5. Manufacturers Cage Code
6. Date of Manufacture
7. Model Number
8. Serial Number
9. Refrigerant Type
10. Refrigerant Quantity
11. Power Requirements, Volts, Amps Watts
12. Refrigeration Unit Weight
13. Technical Manual Number

4.5.14 Reliability Test. A reliability test shall be run as follows:

A minimum of three TRCS units shall each be tested for a minimum of 1600 hours. The individual TRCS unit test hours will be combined and will be the basis for determination of compliance with 3.4.14. Any of the following combinations of total test hours and allowable failures will demonstrate the MTBHEFF of 1600 hours at the 80% lower confidence limit.

Total Combined Test Hours for 3 Test Units	Maximum Allowable Essential Function Failures
4790	1
6846	2
8824	3

Operating hours for the units shall be considered the system on time. The defrost control shall be set for two defrosts in a 24-hour period of operating time.

The refrigeration unit shall be mounted onto the TRCS container or clamped to the evaporator opening of the calorimeter with a similar Heat Leakage Coefficient or the TRCS Insulated Container itself with a minimum volume of 270 cubic feet.

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The air entering the condenser and surrounding the condensing section shall be maintained at a minimum of 100°F for 1200 hours of operation and at a minimum of 125°F (+0/-3°F) for 400 hours of operation.

The refrigeration unit shall be operated with the thermostat set at 0°F for 36 hours and at 35°F for the next 36 hours. This cycle shall be repeated until at least 1600 operating hours have been completed on each test unit. During the test, recorded data must include suction and discharge pressures, air temperature entering the condenser, refrigerated enclosure temperature, amperage, voltage and compressor operating hours every 10 minutes on automated recording instruments.

Failure of the unit to pull down and maintain the set point temperature of the enclosure, failure to start and operate properly, failure to maintain set point temperatures, defrost, remove condensate, or any mechanical or control failure that cannot be repaired by the operator in less than 30 minutes shall be defined as an essential function failure (EFF). A malfunction that can be repaired in less than 30 minutes (by the operator) shall be classified and recorded as an incident.

When failures occur, the failure shall be analyzed, correction made and the test resumed. All failures, analyses, data, test personnel and corrective actions shall be fully recorded in the Test Incident Report.

The test shall be successfully completed when the total combined test time has been completed with no more than the allowable number of Essential Function Failures (EFF) occurring. If more than the allowable number of EFF's occur (see table above), the requirement has not been met and additional test data/time may need to be collected/accrued. Reliability data must include hours of operation, via electric standby operation for type I and III and from the integrated multi fuel engine for Type II.

4.5.14.1 Service and Maintenance Simulation and Inspection. A simulated demonstration shall be accomplished verifying that all major refrigeration unit components are accessible for adjustment, maintenance, servicing and replacement without the removal of any other major component or need for special tools. The simulation shall also illustrate that all other components and assemblies are accessible for maintenance, repair, service or replacement and require no special tools for those actions. Verification shall be made that the moisture content of the refrigeration system can be visually determined. Demonstrate the median time to repair (MedTTR) for all essential unscheduled maintenance demands (EUMD) does not exceed 1 hour. Demonstrate the maximum time to repair (MaxTTR) for 90 percent of all EUMD does not exceed 1.5 hours. Verify that the maintenance ratio (MR) does not exceed .01 maintenance man-hours/operating hours.

4.5.15 RESERVED

4.5.16 Electrical System Wiring and Refrigeration Component Certifications. Wiring practices used in the refrigeration units shall be certified as conforming to National Electrical Code (NFPA) Standard No. 70 by submission of the manufacturer's certification, a certified test report or appropriate labels or listing marks.

Refrigerant containing components used in the refrigeration units shall be certified as conforming to UL Standard 207 and limit controls used shall be certified as conforming to UL Standard 353. Verification of certification shall be conducted via submission of the manufacturer's certification, a certified test report, appropriate labeling or listing marks (by submission of the manufacturer's certification), a certified test report, or other appropriate labeling/listing marks.

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4.5.17 Wire marking inspection. Electrical wiring shall be marked to allow identification with the wiring diagram and wiring schematics located in the Technical Manual.

4.5.18 Transportation Certification. The contractor shall certify compliance with the transportability requirements of 3.4.16 and respective subsections.

4.5.19 Internal Air Transport Certification. The contractor shall be responsible for receiving certification that the TRCS is transportable in C-130 and larger Air Mobility Command (AMC) aircraft (see 6.4.1) without damage to the TRCS or the transporting aircraft, injury to personnel, or operational failure during and after flight. The contractor shall perform, and provide results of, all necessary substantiating analysis, tests, and documentation to support Internal Air Transport Certification efforts to ensure that the TRCS is transportable by USAF fixed wing aircraft. The TRCS shall undergo and pass air transportability testing, if deemed necessary by ATTLA to receive certification. The TRCS shall be subjected to MIL-HNDBK-1791, paragraph 5.2.5.2, Rapid Decompression verification. If required by Air Transportability Test Loading Activity (ATTLA), flight-testing shall first be conducted with the maximum payload uniformly distributed over the entire usable floor space of a single TRCS in the shipping configuration. Only ATTLA may determine whether to allow substantiation by comparative analysis for certification based upon similarity to a previously certified container.

All substantiating structural and transportability data including dimensioned drawings, weight and center of gravity, shall be supplied to the Air Transportability Test Loading Activity (ATTLA), ASC/ENFC Aerial Delivery Group, 2530 Loop Rd West Building 560, Wright-Patterson Air Force Base, OH 45433-7101. Any failure of the TRCS to withstand the stresses induced during the air mode test shall constitute failure of this test event.

For Type II only, verify that a five gallon capacity polyethylene fuel can, complete with mounting hardware, fuel hoses, fitting, and valves are furnished to comply with internal air transportability requirement. The RU shall be operated properly from the 5 gallon fuel for one hour in the cooling mode without failure.

4.5.20 External Air Transport Certification. The contractor shall test and certify compliance per paragraph 3.4.18. The contractor shall provide all necessary support for External Air Transport Certification efforts in conjunction with approved DoD certification agency to ensure that the TRCS is transportable by DOD Rotary winged aircraft (see 6.4.2). All relevant structural and transportability data including weight and center of gravity shall be supplied by the contractor to the US Army Natick Soldier Research Development and Engineering Center, Delivery Engineering Support Team (ADEST), AMSSB-RAD-D(N), Aerial Delivery Group, 15 Kansas Street, Natick, MA 01760. All testing shall be conducted in accordance with MIL-STD-913 and MIL-STD-209. Slinging and tie down provisions for all modes of transportation shall meet the requirements of MIL-STD-209. A failure of the TRCS to meet the measurements needed for successful completion of slinging and tie down testing shall constitute failure of this test event.

4.5.20.1 Static Lift Test. A crane shall lift the TRCS, using an Army Helicopter Sling set; with a range of weights and center of gravities allowed by Army Rotary winged aircraft to include unloaded and fully loaded TRCS. Static lift testing is required to verify rigging procedures, chain link counts and to identify any potential sling interference issues. The TRCS and the lift provisions shall show no visible signs of damage, deformation or performance degradation. This test must be completed first prior to

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performing the proof load test in order to determine the correct lengths of each lifting leg used in the proof load test.

4.5.20.2 Proof Load Test. A crane shall lift the TRCS, using wire rope or chain for a time interval of 90 seconds with a gross TRCS weight of 56,438 lbs or a force of 3.2G times the maximum Army helicopter payload lift weight. A load cell can be used to verify that the appropriate force is applied during this test. In rare instances, tests conducted on identical items, or tests done for other governmental agencies may be used in lieu of required HSL specific tests. The TRCS and the lift provisions shall show no visible signs of damage, deformation or performance degradation.

4.5.20.3 Flight Evaluation Test. The TRCS shall undergo and pass flight evaluation testing, if deemed necessary by ADEST. Actual helicopter flight evaluation tests may or may not be required for some Helicopter Sling Loads (HSL). ADEST is the only facility that has data to allow comparative analysis leading to HSL certification based upon similarity. The TRCS and the lift provisions shall show no visible signs of damage, deformation or performance degradation.

4.5.20.4 Transportation Induced Shock (Drop Test). The TRCS shall be tested and pass the 6-in. transit drop test in accordance with MIL-STD-810G, Method 516.6, Procedure IV-Transit Drop and per paragraph 3.4.22 at the Maximum Gross Weight. The TRCS container shall be coupled to 2 other dry freight TriCons and tested in the 1C configuration in the outboard position. The contractor shall verify that the TRCS, does not exhibit any physical damage, permanent deformation, compromise, detriment in operation of the TRCS, its cargo or restraint systems, buckling, delamination, seal separation, system failure or structural failures of any parts constitutes failure of this test. The TRCS will not be required to operate when the system is being dropped. Upon test completion, the TRCS shall be operated and required to maintain internal set point temperature conditions at thermostat settings of 0ºF and 35ºF for a period of at least 2 hours without any type of failure. The TRCS shall be instrumented to collect and record acceleration and G-force data.

4.5.21 Rail Impact Test (transport mode). Perform a rail impact test with three TRCS coupled together and/or a combination of TRCS and dry freight TriCons as a 1C container in accordance with MIL-STD-810, Method 516.5, Procedure VII (rail impact) with the TRCS at its maximum gross weight. The TRCS shall meet the Gabarit International de Chargement (GIC) equipment gauge envelope as defined in MIL-STD-1366. The payload shall be secured inside the TRCS. No external blocking, bracing or dunnage is allowed. Verify that only the internal cargo restraint system (see 3.2.24) is utilized to secure internal payload and that no adjustments are made to these provisions during the test. The TRCS will not be required to operate throughout the entire test duration of rail impact testing. Secure the TRCS to a Container on Flat Car (COFC) type rail car utilizing only the ISO corner locks furnished with the flatcar. No adjustments of the ISO fittings or locking devices are allowed following test commencement.

Conduct a visual inspection before and after each impact and perform a operational checkout, prior to, and after the Rail Impact test to ensure there is no physical damage, permanent deformation, compromise, detriment in operation of the TRCS, its cargo or restraint systems. This also includes any buckling, delamination, seal separation, system failure, structural failure or failure of any other type or part. Any instances of the previously listed items, constitutes failure of this test. Upon test completion, the TRCS shall be operated, and required to maintain internal set point temperature conditions at thermostat settings 35 ± 2°F for a period of at least 1 hour. The TRCS shall be instrumented to collect and record acceleration and G-force data.

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4.5.22 Ground Mobility (transport mode on PLS truck and trailer). The TRCS, when fully operational and mounted on a prime mover, shall be capable of transportation over primary roads, secondary roads, and cross-country terrain as specified in TABLE IV. The TRCS, with a simulated payload shall be tested in accordance with MIL-STD-810 Method 514.5 Procedure III and Annex A Category 6 Truck/trailer/tracked large assembly cargo with the exception that all of the parameters from TABLE IV are tested. Verify that no additional external blocking, bracing, dunnage, tie down provisions are utilized to secure the TRCS during this test. Verify that only the internal cargo restraint system is utilized to secure internal payloads and that no adjustments are made to these provisions during the test. Conduct a visual inspection, and an operational checkout, prior to, and after, the Ground Mobility testing. Upon completion of Ground Mobility testing, verify by inspection/operation there are no physical damage, permanent deformation, compromise and detriment in operation of the TRCS, its cargo or restraint systems, or prime mover. This also includes; buckling, delamination, seal separation, system failure or structural failures of any parts. Any of the previously listed items constitutes failure of this test. Validation of the ability of the TRCS to fit on Army motor transport vehicles shall be accomplished by verifying the dimensions, weight, and adherence to ISO criteria for freight containers. Successful attainment of CSC certification is required before conducting any ground mobility testing. Mileages and courses are as specified below:

Table IV Ground Mobility

- Limited mileage ((at the empty (TARE weight) and maximum payload configuration)) -

Surface/Test Course	Prime Mover Mileage				Max. Speed (MPH)	Average Speed (MPH)
Prime Mover:	PLS Truck and Trailer					
TRCS Gross Weight (pounds)						
Highways - paved roads (miles) (Perryman straightaway)	250 miles empty 250 miles fully loaded				60	50
Gravel roads (miles) (Munson gravel road course)	650/650 Miles				20	10
Cross-country (unimproved) (Perryman cross- country ((secondary road "A"))	100/100 miles				20	10

The TRCS shall be instrumented to collect and record acceleration and G-force data.

4.6 Environmental Requirements Verification. All environmental requirements shall be verified as follows:

4.6.1 Temperature Certification. In addition to the testing listed herein, the contractor shall provide certification that the TRCS will set up, tear down, operate and function properly, in all ambient

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conditions of both hot and basic climatic categories as defined in AR 70-38. These are to include subcategories of Hot-Dry, Hot-Humid, Constant High Humidity, Variable High Humidity, Basic Hot and Basic Cold Environments. This is to account for all contributing factors of ambient air temperatures, solar radiation, ambient relative humidity's, induced air temperatures and humidity's. Also, it is to account for their respective combinations as outlined in the document.

4.6.2 Weatherproofness Test. Weatherproofness shall be tested in accordance with ISO 1496-3, Test No. 12 – Weatherproofness. Upon completion of testing, verify that no water has leaked and/or penetrated into the container and the refrigeration/heating unit shall function properly.

4.6.3 Salt Fog Test. Expose the TRCS to a marine atmosphere and test in accordance Method 509.4 of MIL-STD-810, except that the salt solution shall have a concentration of 10%. Ensure that the Salt Fog Test is conducted just prior to the Fungus Test. The 96 hour period shall be conducted as follows: First 24 hours of exposure, second 24 hours of drying time, third 24 hours of exposure and lastly 24 hours of drying time. Drying period will require an ambient temperature of at least 70±5°F and a relative humidity of 50%, or less. Note: The TRCS does not have to operate throughout this testing. Immediately following completion of the 96 hour testing, turn the system on and adjust the conditions to maintain internal set point temperature conditions at thermostat settings of 0ºF and 35ºF (±1°F). Allow the TRCS to operate for a period of at least two hours and record results of testing. Verify that the system is constructed of compatible materials, is inherently corrosion resistant and treated to provide protection against corrosion and deterioration in both operational and storage scenarios. Verify that the TRCS operates properly and that there is no evidence of corrosion, rust, damage, degradation or any other form of deterioration. Ensure that no dissimilar metals have been used in intimate contact with one another unless protected from galvanic corrosion.

4.6.4 Humidity Test. Perform a humidity test on the TRCS in accordance with Method 507.4, MIL-STD-810. The test will consist of a 24-hour conditioning period, immediately followed by a minimum quantity of five 48-hour cycles as specified in Method 507.4 of MIL-STD-810. After the 5 cycles, turn the TRCS on and adjust the conditions to maintain internal set point temperature conditions at thermostat setting of 35ºF (±2°F). Allow the TRCS to operate for a period of at least four hours and record results of testing. Verify that the TRCS can withstand this test without evidence of corrosion, swelling, structural deterioration, degradation, property alterations, permanent deformation, physical damage, compromise, buckling, delamination, seal separation, sealing property degradation, operational failures, operational detriment, control circuit damage, without alteration in required clearances, without power supply damage, ensuring proper system capacity, without discoloration, dimensional change, melting, damage to lubrication systems and other related effects, without skin failure, system failure or failures of any other type. Any deviation in parameters, as specified, constitutes a failure. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report.

4.6.5 Blowing Sand and Dust Test (operational mode). The TRCS shall be tested for sand and dust in accordance with Procedure I (Blowing Dust) and Procedure II (Blowing Sand) of Method 510.4, MIL-STD-810 to ensure that the TRCS is not affected in any way. TRCS shall be operated and maintain internal set point temperature conditions at thermostat settings of 35ºF (±2ºF) during the last ten minutes of exposure to the most vulnerable face for each procedure. Additionally, the TRCS shall be operated and maintain internal set point temperature conditions at thermostat setting of 35ºF (±2ºF) for a period of two hours following completion of Sand and Dust Testing. Demonstrate that the TRCS remains operable, as intended, throughout the duration of this test and is in no way damaged in sand and

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dust conditions. Verify that the TRCS does not exhibit any physical damage, deformation, compromise, detriment in operation, delamination, any skin/paint damage, seal separation, loss of sealing properties, damage to control circuits, alteration in required clearances/dimensions, damage to power supplies, reduction in system capacities, damage to lubrication systems and other related effects, system failure or failures of any other type. Any deviation in parameters, as specified, constitutes a failure. Any infiltration of sand/dust into the food storage areas, the evaporator section or any degradation in performance of the TRCS by sand/dust that cannot be corrected by routine maintenance by the TRCS operator constitutes a failure. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report.

4.6.6 Fungus Test. The contractor shall provide certification that shows that the TRCS is fabricated of materials that are resistant to the growth of fungi and that they do not contain any fungus sustaining nutrients.

4.6.7 Transportation Operation Certification (NOT REQUIRED FOR TRCS CONTRACT). Ensure that the TRCS possesses transportation certification that it will not be affected/ hindered (in any way) by any type of environmental conditions and operate properly in all potential scenarios. This is to include, but not limited to, humidity, rain, freezing rain, snow, ice, sand, dust, elevation worst case ambient conditions (AR 70-38 hot and basic climates), high temperature and humidity, as well as low temperature and humidity.

4.6.8 Storage Test and Certification. The TRCS shall be subjected to a temperature of 160°F for 4 hours, then the room temperature shall be brought to 70°F $\pm$ 5°F and the unit operated for 4 hours. The refrigeration unit shall then be subjected to a temperature of -60°F for 4 hours, then the room temperature shall be increased to 50°F $\pm$ 5 °F and the unit operated for 2 hours at a thermostat set point of 35°F and then another 2 hours at 0°F. Verify that the TRCS does not exhibit any physical damage, permanent deformation, compromise, detriment in operation, buckling, delamination, seal separation, loss of sealing properties, damage to control circuits, any alteration in required clearances/dimensions, damage to power supplies, detriment in system capacities, discoloration, melting, damage to lubrication systems and other related effects, system failure or failures of any other type as a result of being exposed to these storage environments. The contractor shall provide certification that the TRCS will perform its intended functions properly and safely after being subjected to storage in all ambient conditions of both hot and basic climatic categories as defined in AR 70-38. These are to include subcategories of Hot-Dry, Hot-Humid, Constant High Humidity, Variable High Humidity, Basic Hot and Basic Cold Environments. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report.

4.6.9 Noise Level Test. The TRCS unit shall be tested to meet the noise limits as specified in MIL-STD-1474 D. The noise at the operators and servicing positions shall not damage personnel hearing while operating. The noise level shall not exceed 85dB. For Type II, the RU shall operate from the integrated multi fuel engine. A certified test report shall be provided to the Government for verification

4.6.10 Solar Radiation (Sunshine) Test. A solar radiation test shall be performed in accordance with Procedure I (Cycling) Method 505.4 of MIL-STD 810 under the Climatic Design Type of A1 (Hot-Dry reference Figure 505.4-1 Procedure I Cycling Time). Verify that Procedure II Type lamps are utilized to provide some limited actinic effects. Verify that the TRCS operates properly throughout the entire duration of this test (See Table VI listed below). Testing will be conducted on a green container. Under Procedure I, Step 2, conduct a total of four (4) 24-hour cycles. Verify that the TRCS can

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withstand this test without evidence of corrosion, swelling, structural deterioration, degradation, any operational failures, property alterations, permanent deformation, physical damage, compromise, detriment in operation, buckling, delamination, seal separation, loss of sealing properties, damage to control circuits, alteration in required clearances/dimensions, damage to power supplies, detriment in system capacities, discoloration, melting, damage to lubrication systems and other related effects, exhibit in any skin failure, system failure or failures of any other type. Any deviation in parameters, as specified, constitutes a failure. The TRCS shall be instrumented to collect and record time, temperature and wattage per square meter data that shall be graphed in the FAT Report.

4.6.11 Finishing, Cleaning, Treating, and Painting inspection. Except where detrimental to proper operation, verify that external surfaces of the TRCS are cleaned, treated, primed, and top coated. Painted surfaces shall be visually examined to verify that there is no streaking, cropping, pin-holed surfaces, checking, cracking or missed areas.

4.6.12 Ice and Freezing Rain. An ice and freezing rain test shall be performed in accordance with MIL-STD-810, Method 521.2 utilizing glaze ice with a cumulative thickness of .512 inches (13 mm) and a wind velocity of 0 m/hr. The ice is to be applied to the top surface and on all four sides of the TRCS. Verify that the TRCS remains operational throughout the entire duration of testing and maintains internal set point temperature conditions at thermostat setting 35ºF (±1ºF). Verify the removal of ice, following the test, results in no damage to the TRCS. Ice is to be removed via typical ice removal devices/procedures utilized by Warfighters. Demonstrate that the TRCS does not allow the accumulation of water in such a way as to cause failure or bind components upon freezing. Verify the TRCS shows no sign of physical damage, degradation in performance/operation, that the ice/freezing rain does not bind or damage moving parts, add significant weight, increase hazards/failures, reduce efficiency or is affected in any way by ice, snow, and/or freezing rain as a result of this test. Verify that the TRCS can withstand this test without evidence of corrosion, shrinkage, structural deterioration, degradation, any operational failures, property alterations, permanent deformation, physical damage, compromise, detriment in operation, buckling, delamination, seal separation, loss of sealing properties, damage to control circuits, alteration in required clearances/dimensions, damage to power supplies, detriment in system capacities, discoloration, dimensional changes, cracking, damage to lubrication systems and other related effects, exhibit any skin failure, system failure or failures of any other type. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report.

4.6.13 High Temperature Test. The TRCS shall be tested in accordance with test method standard MIL-STD-810, Method 501.4, Procedure I (Storage) and Procedure II (Operation).

Procedure I (Storage): Subject the TRCS to seven (7) 24-hour cycles of climatic design type A1 as listed in TABLE 501.4-II. High temperature cycles, climatic category - Hot under the induced temperature and humidity conditions. Immediately following completion of all cycles, turn the system "ON", and adjust the conditions to maintain internal set point temperature conditions at thermostat settings of 0ºF and 35ºF (±1ºF). Allow the TRCS to operate for a period of at least four hours and record results of testing.

Procedure II (Operation): Conduct per MIL-STD-810F. During these 6-hour time intervals, ensure that the refrigeration systems compressor operates for at least 70% (±5%) of the time and maintains an internal TRCS temperature of 0ºF (±1ºF), and operates properly, throughout this portion of the test. Verify that the TRCS operates continuously and maintains internal set point temperature conditions as specified below in Table VII A.

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Table VII A (Multi Fuel Engine Drive Operation, Type II)

Document:	Ambient Air Condition Reference:	Climatic Design Type:	Classification:	Cycle Quantity:	Configuration:	Internal Temperature Tolerance:
MIL-STD 810	Table 501.4-II	Hot	A1 (Hot-Dry)	1	Single Temp 0ºF	+/-3ºF
AR 70-38	Figure Table 2-3	Hot	B3 (Hot-Humid)	1	Single Temp 0ºF	+/-3ºF
AR 70-38	Figure Table 2-4	Basic	B1 (Constant High Humidity)	1	Single Temp 0ºF	+/-3ºF

Table VII B (Electrical Operation, Type I and Type III)

Ambient Air Condition Reference:	Climatic Design Type and Classificaton:	Frequency: Tolerance (+/- 3.5%)	Power Source: Tolerance (+/- 10%)	Time Period: (hours)	Configuration:	Internal Temperature Tolerance:
Table 501.4-II	A1 (Hot-Dry)	60-Hz	208 Volt 3-phase	6	Single Temp 0ºF	+/-3ºF
MIL-STD-810	125ºF	60-Hz	208 Volt 3-phase	6	Single Temp 0ºF	+/-3ºF

Verify that the TRCS can withstand this test without evidence of corrosion, swelling, structural deterioration, degradation, operational failures, property alterations, permanent deformation, physical damage, compromise, without detriment in operation, buckling, delamination, seal separation, loss of sealing properties, damage to control circuits, alteration in required clearances/dimensions, damage to power supplies, detriment in system capacities, discoloration, melting, damage to lubrication systems and other related effects, does not exhibit any skin failure, system failure or failures of any other type. Verify the TRCS operates properly and is able to maintain set point and product temperatures. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report. Any deviation in parameters, as specified, constitutes a failure.

4.6.14 Low Temperature Test. The TRCS shall be tested in accordance with test method standard MIL-STD-810, Method 502.4 Procedure I (Storage) and Procedure II (Operation). Conduct Procedure I prior to Procedure II. Return the chamber temperature to at least 60ºF and verify that no damage has occurred, as outlined below, immediately following each procedure. All Low Temperature Testing is to be conducted while utilizing 208VAC, three phase, 60 Hz power or the integral multi fuel engine drive.

Procedure I (Storage): Subject the TRCS to two (2) 24-hour periods of the lowest temperature ambient (-38ºF from climatic design type C1 (Basic-Cold) as listed in Figure TABLE 2-7 (from AR 70-38). Immediately following completion of all cycles, and allowing the TRCS to return to -35ºF, turn the system on and adjust the conditions to maintain internal set point temperature conditions at thermostat settings of 0ºF and 35ºF (±1ºF). Allow the TRCS to operate for a period of at least two hours and record results of testing.

Procedure II (Operation): The TRCS shall be subjected to a total of five (5) 24-hour cycles as specified below in Table VIII A and/or Table B. Verify that the TRCS operates continuously and maintains internal set point temperature conditions as specified. Single temperature operation is to consist of the full interior of the TRCS operating with a thermostat setting of 35ºF.

Table VIII A (Engine Drive Operation, Type II)

Document:	Reference:	Climatic Design Type:	Classification:	Cycle Quantity:	Configuration:	Tolerance:
AR 70-38	Figure Table 2-7	Basic	C1 (Basic Cold)	5	Single Temp 60ºF	+/-3ºF

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Table VIII B (Electrical Operation, Type I and III)

Ambient Air Condition Reference:	Climatic Design Type and Classification:	Frequency: Tolerance (+/- 3.5%)	Power Source: Tolerance (+/- 10%)	Time Period: (hours)	Configuration:	Internal Temperature Tolerance:
AR 70-38 Figure Table 2-7	C1 (Basic Cold)	60-Hz	208 Volt 3-phase	6	Single Temp 60ºF	+/-3ºF

Verify that the TRCS does not exhibit any physical damage, permanent deformation, compromise, detriment in operation, buckling, delamination, seal separation, loss of sealing properties, damage to control circuits, any alteration in required clearances/dimensions, damage to power supplies, detriment in system capacities, discoloration, damage to lubrication systems and other related effects, does not exhibit any system failure or failures of any other type. Verify the TRCS operates properly and is able to maintain set point and product temperatures. The TRCS shall be instrumented to collect and record time and temperature data that shall be graphed in the FAT Report. Any deviation in parameters, as specified, constitutes a failure.

4.6.15 Blowing Rain Test (operational mode). The TRCS shall be tested and pass the blowing rain test in accordance with MIL-STD-810F, Test Method 506.4, Procedures I and II, (40 mph, 4 in/hr). The refrigeration unit shall be mounted on the actual TRCS. A visual verification shall be done to insure no leakage occurred in the evaporator section, electrical control or junction boxes or the food storage area. The refrigeration shall not be adversely affected during or after being subjected to this test.

4.6.16 Electromagnetic compatibility test. The TRCS (both electrical and engine driven systems) shall be tested in accordance with MIL-STD-461. An EMI test report shall be compiled documenting the results of all tests performed. The TRCS shall meet the RE102 and CE102 limits of MIL-STD-461. The TRCS shall be instrumented to collect and record EMI data that shall be graphed in the FAT Report.

4.6.17 Shelving Test and Inspections (Option specified on contract). Verify the presence of a NSF approved (Food Grade Stainless Steel), removable shelving system consisting of five (5) shelves per side, ten (10) shelves total. Verify that each shelf is a minimum of 22” wide and the entire interior width of the container. Test each shelf by placing a 600 pound (distributed load) on it. Verify through testing the unloaded shelving system is capable of withstanding the all of the transportability requirements listed herein without physical damage to the shelving, or the interior of the container. Test the shelving system by placing a 600 pound distributed load on each shelf. No structural deformation or damage to the container is allowable. Verify the shelving system and tie down system is capable of holding ten (6) 95% percentile human cadavers in body bags (3 per side) without contacting one another or falling off the shelves. Verify the presence of an isle way. Verify that the shelving does not cover the IC floor drain opening or hinder IC box drainage and maintainability of the floor drain. Verify the shelving does not hinder accessibility/maintainability of the evaporator section. Demonstrate the shelving system does not require any special tools and designed for easy access to mounting hardware. Verify that each shelf is supplied with captive tie down straps or a netting system to prevent the movement and or damage of the cargo and the TRCS. Verify there a no loose shelving parts. Verify the shelving is designed to promote airflow around the cargo. The shelving system shall also be designed for the collection and pooling of fluids without spilling on the floor or the lower shelves and designed to store Unitized Group Ration boxes without wasted space. Note: Unitized Group Ration boxes are 24” long X 14” inches wide X 9.5” inches high.

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4.7 Demonstration. The contractor shall demonstrate the effective working of the TRCS in the following manner: The RU shall be mounted and un-mounted from the IC to demonstrate that it shall be easily serviced, maintained and replaced in the field by soldiers. Verify that the TRCS and RU do not exhibit any physical damage, compromise, detriment in operation, loss of sealing properties, or failure/damage of any other type following this procedure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's, or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity. All TM's shall come over-packed with the TRCS.

6. NOTES

6.1 Intended Use. The TRCS system is intended for use in the transportation of equipment and storage of non-perishables and perishables items (food) and other temperature sensitive commodities (human remains) requiring controlled temperature. These containers are based on commercial designs, manufactured using commercial processes, and are modified for military unique requirements for operation in an extreme environment, under conditions typically found on the battlefield.

6.2 Acquisition Requirements. Acquisition documents must specify the following:

a. Title, number and date of this purchase description.
b. Issue of ASSIST to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.2.1, and 2.3)
c. When a first article is required (see 3.1, 4.1.1 and 6.3)
d. When a color other than green is required (see 3.5.12).
e. Sampling for conformance inspection (see 4.1.2).
f. Packaging requirements (see 5.1)

6.3 First Article. When requiring first article inspection, contracting documents will provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents will provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products, wishing to rely on such production testing, must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Verification Alternatives. Contracting documents will provide guidance to offerors regarding the submission of alternatives to specified verification methods (see 4.1.4).

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6.4.1 Internal Air Transportation. Certification that the TRCS meets the requirements for movement by fixed wing USAF aircraft will be obtained from the Air Transportability Test Loading Agency - Engineering Directorate (ATTLA), ASC/ENFC, Building 560, 2530 Loop Road West, Wright-Patterson Air Force Base, Ohio 45433-7101 and engineering.wpafb.af.mil/attla/attla.asp

6.4.2 External Air Transportation. Certification that the TRCS meets the requirements for movement by External air rotary wing aircraft will be obtained from the US Army Natick Soldier Research Development and Engineering Center, Aerial Delivery Engineering Support Team (ADEST), AMSSB-RAD-D(N), Aerial Delivery Group, 15 Kansas Street, Natick, MA 01760.

6.4.3 Rail Delivery Certification. Certification that the TRCS meets requirements for movement by rail will be obtained from the Director, Surface Deployment and Distribution Command, Transportation Engineering Agency (SDDC-TEA), 720 Thimble Shoals Road, Suite 130, Newport News, VA 23606-0276.

6.5 Technical Manuals. The requirement for technical manuals will be considered when this performance specification is applied on a contract. The technical manuals will be acquired under a separate contract line item in the contract. At a minimum, the TRCS technical manuals will provide the proper procedures to include but not limited to: specifications, operations, maintenance, troubleshooting, repair, coupling, decoupling, preparation for movement and storage, transportation, etc…

6.6 Ozone-Depleting Substances. A listing of ozone-depleting substances prohibited from use can be found in publication SD-14 Listing of Toxic Chemicals, Hazardous Substances, and Ozone-Depleting Chemicals. It is available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

6.7. Recycled, Recovered, or Environmentally Preferable Materials. Used, rebuilt or remanufactured components, pieces, and parts will not be incorporated into the TRCS. Damaged or repaired items will not be used unless otherwise specifically approved by the procuring activity. Materials not specified herein will be in accordance with Federal, Military, and/or National Technical Societies, Associations or Institute specifications and/or standards.

6.8 Subject Term (key word) Listing.

Electric driven refrigeration unit
ISO container
Refrigerator, field
Storage, food
Transportation, food

6.9 Supersession Data. Not Applicable

6.10 Changes from Previous Issue. Not Applicable

6.11 Except as otherwise specified herein, the container will be the standard product of the contractor. The TRCS will be new and unused. New and unused containers will be sealed at the factory and shipped without cargo to the final destination. The TRCS will consist of the following major components and subassemblies: 1) ISO compliant frame and panel structure, 2) ISO compliant corner castings, 3) forklift pockets, 4) high quality insulation, 5) equipment interface points, 6) door assembly, 7) door latches, 8)

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door gasketing/seals and 9) food grade interior and flooring, mechanical refrigeration unit, condensate drain assembly, ISO interconnectors, etc…

6.12 Material. The materials selected for construction of the containers are the responsibility of the contractor and will be in accordance with an applicable Federal, Military or National Technical Society, Association or Institute specification or standard. The materials will be of sufficient durability to meet all the performance requirements as specified herein. The material will have no adverse effect on the health of personnel when used for its intended purpose. Toxic chemicals, hazardous substances, or ozone depleting chemicals (ODC) will not be used. No radioactive materials, mercury, or cadmium plating will be used. Care will be exercised during design and in all material selection to reduce or eliminate the need for hazardous materials and emissions of Volatile Organic Compounds (VOCs) throughout the product's lifecycle.

6.13 Metal Fabrication. Metal used in the fabrication of equipment will be free from kinks and sharp bends. The straightening of material will be done by methods that will not cause injury to the metal. Shearing and clipping will be done neatly and accurately. Corners will be square and true. Flame cutting, using a tip suitable for the thickness of the metal, may be employed instead of shaving or sawing. Burned surfaces or flame-cut materials will be free of slag. All bends of a major character will be made with controlled means in order to insure uniformity of size and shape. Precautions will be taken to avoid overheating, and heated metal will be allowed to cool slowly. Warped metal plates and oil canning of panels will not be acceptable.

6.14 Bolted and Riveted Connections. Bolt and rivet holes will be accurately punched or drilled and will have the burrs removed. Washers, lock washers, or lock nuts will be provided where necessary and are preferred over the use of rivets. All bolts, nuts, and screws will be tightly secured and the sizing will be standardized to the maximum extent possible. All components that are prone to physical damage, will be designed to provide easy access to the bolt heads allowing the use of common hand tools. When necessary, rivet heads, when not countersunk or flattened, will be uniform in size and shape for the same diameter or rivet concentric with the rivet holes, and in full contact with the surface of the member. Corrosion resistant hardware is required. Threaded inserts or machine screws are required where sheet metal parts are joined together. Water tightness must be maintained where pop rivets are used to attach components or plates to the container.

6.15 Fire Safety. The Supplier will minimize the potential for fire damage to buyer equipment through selection of fire retardant construction materials. Such materials include, but are not limited to, wall, roof and floor construction materials, wall ceiling and floor coverings and finishes. Materials selected will be listed by a recognized agency (such as UL) or certified by the manufacturer to comply with recognized standards (such as ASTM). The supplier will use materials and products having a rating of Class 1 and flame spread ASTM rating less than 25, or have such material reviewed and approved by the buyer.

6.16 Material Deterioration, Prevention, and Control. The containers will be fabricated from materials, inherently corrosion resistant such as aluminum, muffler grade stainless steel, CORTEN steel, galvanized steel to G-90 level etc… The container will be use treated materials that provide protection against the various forms of corrosion and deterioration that may be encountered in all of the operating and storage environments addressed in Section 3.0. The containers will operate for a minimum of 20-year service life without performance degradation due to corrosion. No actions beyond normal washing, scheduled maintenance (exclusive of paint touch up), repair of accidentally damaged areas (not a result of intended use, deficiency in design, materials, manufacturing or normal wear), and replacement of damaged paint will be necessary to keep the corrosion prevention in effect. To meet the Nuclear Biological and Chemical (NBC) survivability requirements, materials will be selected which can be decontaminated without degradation when cleaned with DS2. Materials used will be non-nutrient to

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fungus. The connecting couplers will be considered part of the container and will also meet the 20-year service life; regardless that normal use of couplers.

6.17 Dissimilar Metals. Dissimilar metals will not be used in intimate contact with each other unless protected against galvanic corrosion. Coatings for purposes of electrolytic protection will not be exposed to abrasion where coatings would degrade. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

6.18 Welding. Welding will be done in accordance with AWS D1.1, AWS D1.2, and/or ASME Boiler and Pressure Vessel Code, as applicable. The surface of parts to be welded will be free from rust, scale, paint, grease or other material. Weld penetration will be such as to provide transference of maximum design stress through the base metal juncture.

6.19 Brazing. Brazing will be done in accordance with AWS B2.2 or ASME Boiler and Pressure Vessel Code, as applicable.

6.20 Workmanship and Construction. All workmanship will be in accordance with engineering, manufacturing, and production standards of the industry as applicable. The TRCS will be free from workmanship deficiencies that could impair its operation or serviceability. All parts components and assemblies, including but not limited to, bearings, seals, sealing agents, machinery, stampings, welded parts, castings, forgings, molded parts and machined work/surfaces will be clean and free from sand, dirt, fins, pits, scales, flux and other harmful extraneous material. Any components and assemblies, which have been repaired or modified to overcome deficiencies, will not be used without prior specific approval of the contracting officer. External surfaces will be free of burrs, sharp edges, and corners, except when sharp edges and corners are functional. The internal cargo space will be free from sharp protrusions that could damage cargo or personnel. The container will be constructed to minimize recesses and voids where moisture can accumulate. No part of the container will protrude beyond the outside surfaces of the corner fittings. Metal used in the fabrication of the container will be free from kinks and sharp bends. Shearing and punching will be done neatly and accurately. Corners will be square and true. All bends of a major character should be made with controlled means in order to prevent degradation of the material and to insure uniformity of size and shape. Castings will be sound and free from patching, misplaced coring, warping or defects, which might render castings unsound for use.

6.21 Test Data. The following tests require automatic data acquisition capabilities to capture operational test data to include, but not limited to, set-point temperatures, actual internal box temperatures, ambient temperatures, evaporator and condenser air inlet and outlet temperatures, voltages, amperage, low and high side pressures, wattage (power), air velocity, humidity, Refrigeration Capacity (-0ºF and 35ºF), Heating Capacity, Power, Airflow, Heat Leakage, Auxiliary fuel tank, Humidity, Salt Fog, Sand/Dust, EMI, Transportation, Solar Radiation, Ice and Freezing Rain, High Temperature, Low Temperature, Reliability, etc…

6.22 Corrosion Control Methods. Corrosion control can be achieved by a combination of design features (as in TACOM Design Guidelines for Prevention of Corrosion in Combat and Tactical Vehicles, March 1988), material selection (e.g. composites, corrosion resistant metal, galvanized steel), organic or inorganic coatings (e.g. zinc phosphate pre-treatment, corrosion resistant plating, E-coat, powder coating) and production techniques (e.g. coil coating, process controls, inspection and documentation). To obtain high level of corrosion resistance/paint adhesion, the manufacturer may want to consider the use of new high performance, "Combat Grade" Powder Coat Primers. These primers can only be applied by approved applicators. The Specification controlling both the primers and their application are available upon request from TACOM - AMSTA-TR-E/MEPS, (586) 574-5083.

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6.23 Warranty. The structural, finish, marking and quality control requirements will be outlined in the solicitation.

6.3 Definitions.

6.3.1 Dissimilar metals. For its purpose, this document considers metals dissimilar when two specimens in contact with each other promote accelerated galvanic corrosion (see 3.2.3).

6.3.2 Recovered materials. For the purpose of this document, recovered materials are those materials that have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials (see 3.2.4).

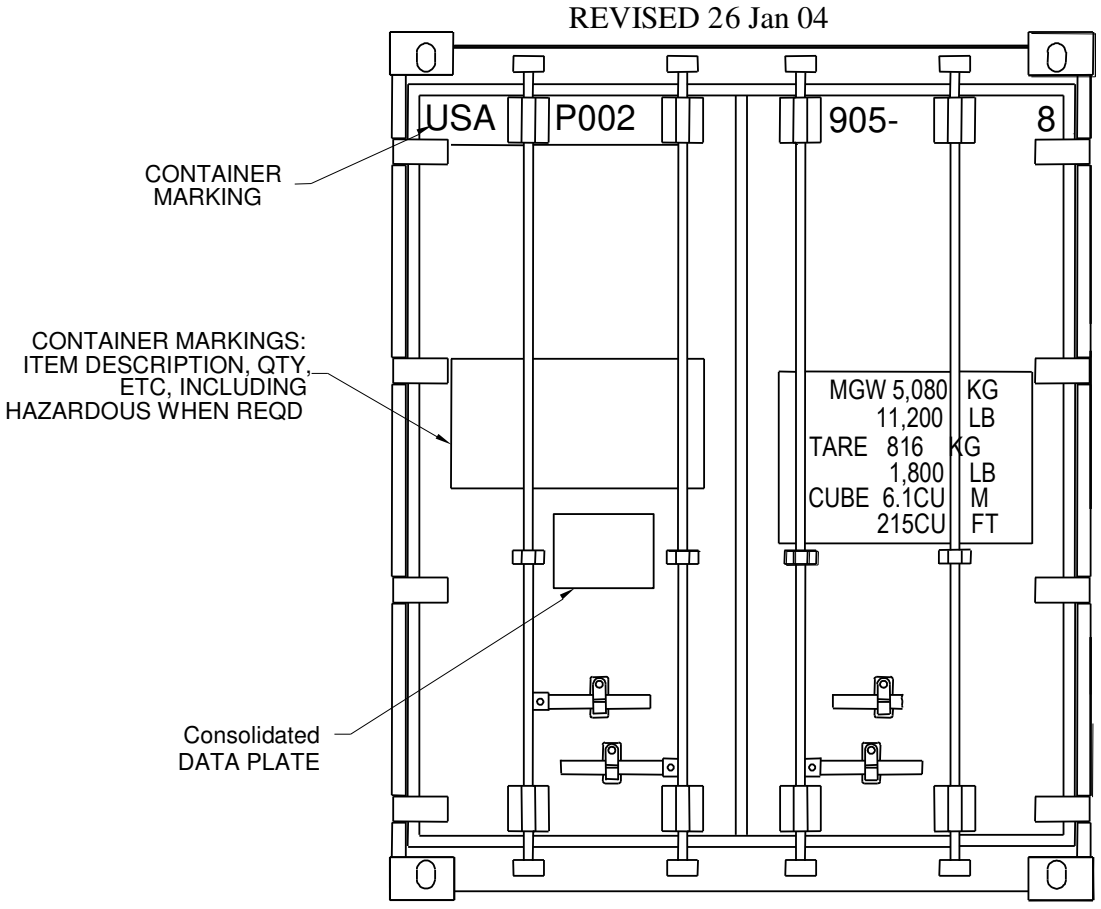


FIGURE 1. Representative TRCS markings.

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Custodian:
Army - GL
AF-99

Preparing activity:
Army - GL
(Project No. 4110-2010-003)

Review activity:
AF-93, 84, 50, 35

DLA – IS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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